SNA – advance topics and case studies

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March 24 (rescheduled), April 1, 2026 (mass bunk),

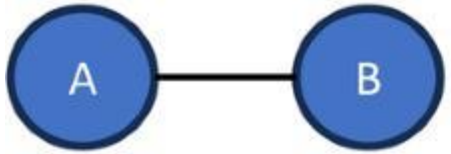
April 8, 2026 (election training)

Key 2025-2026 Emerging Topics and Articles - SNA

1. **Causal Network Analysis**: **Causal inference** within network structures. < directly relevant to modern AI, epidemiology, and platform economics>
2. Methodological Advances: **Deep link entropy** for edge significance and **FONDUE framework** for node disambiguation/deduplication.
3. **Network Applications**: Studies on **collaborative dynamics** using ERGM approaches (e.g., in research partnerships) and analysis of Wikipedia as a **cultural network**.
4. Health & Sociology: **Population-scale network mapping and network interventions** for communities
5. Online & Social Media: **Information diffusion modelling based on explosion shock wave theory** and analysis of non-governmental organization (NGO) interest groups on Facebook.

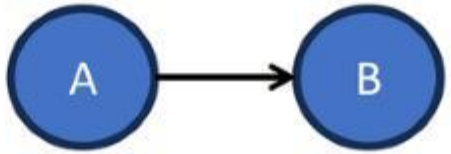
CNA

- **Causal Network Analysis** in Social Network Analysis studies how causal effects propagate through a network where units influence each other.
- It asks not just *“who is connected to whom?”*
  but *“who causes what, through which network pathways?”*



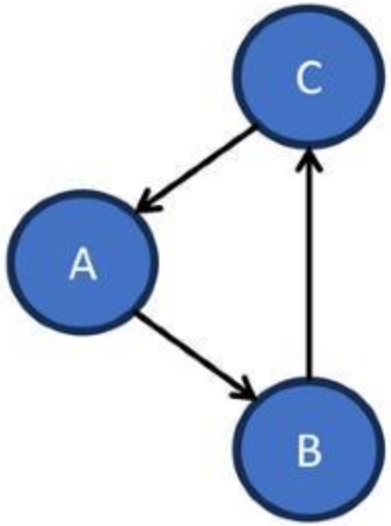
A is adjacent to B and B is adjacent to A; Undirected Link

Undirected Graph



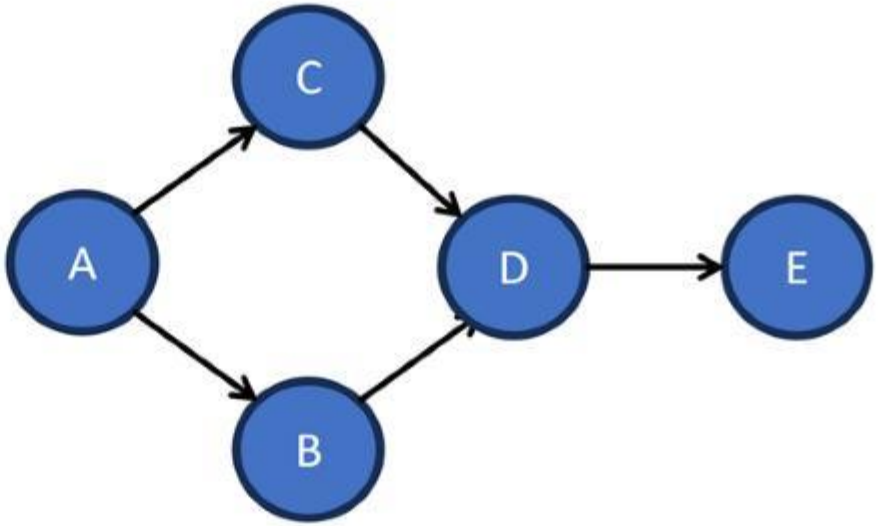
A is adjacent to B but B is not adjacent to A; Directed Link

Directed Graph



A is a causal link to B, B which is a causal link to C and C cycles back to A

Cyclic Graph

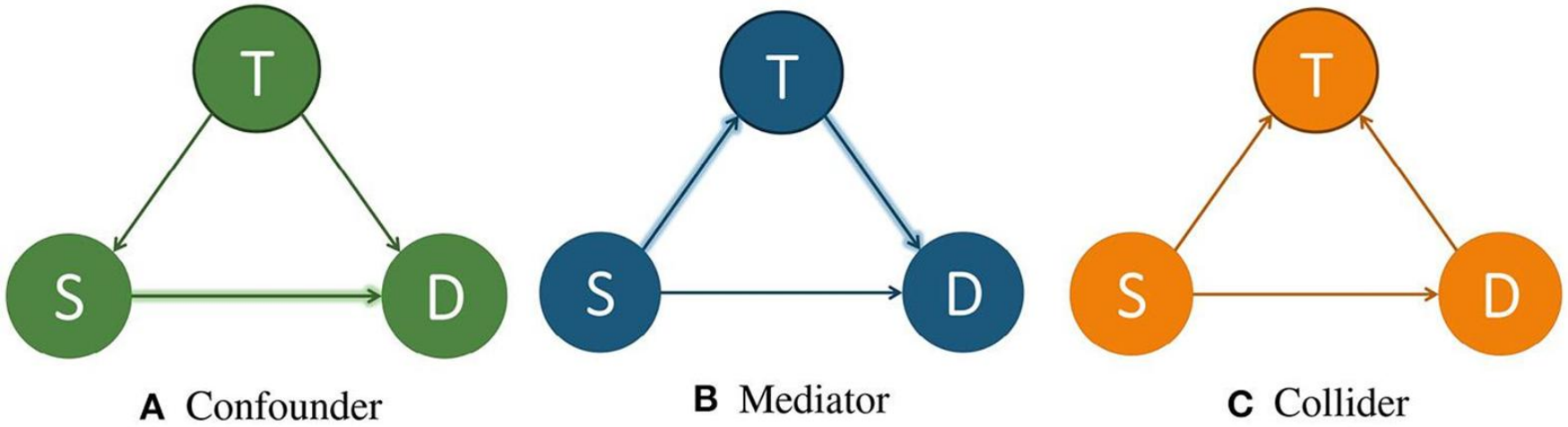


A is a causal link to B and C, which are causal links to D; D leads to the directed causal outcome E

Directed Acyclic Graph

Causal Network Analysis

- In networks, **causality is not local—it is relational**.
- **(1) Confounding + homophily**
- People connect because they are similar
- Hard to separate:
  - **influence vs selection**



Applications – Causal Network Analysis

(A) Epidemiology

- Disease spread
- Vaccination strategies

(B) Social media

- Information diffusion
- Fake news propagation

(C) Economics

- Peer effects in education
- Financial contagion

(D) Marketing

- Viral campaigns
- Influencer targeting

Aspect	SNA	Causal Network Analysis
Focus	Who is connected	Who influences outcomes
Nature	Descriptive	Causal
Output	Patterns	Effects

Intuition

- Vaccinating person A affects:
- A's outcome (direct effect)
- B and C (neighbors → spillover effect)

Case Study: Causal Inference in Networked Epidemics (Vaccination & Spillovers)

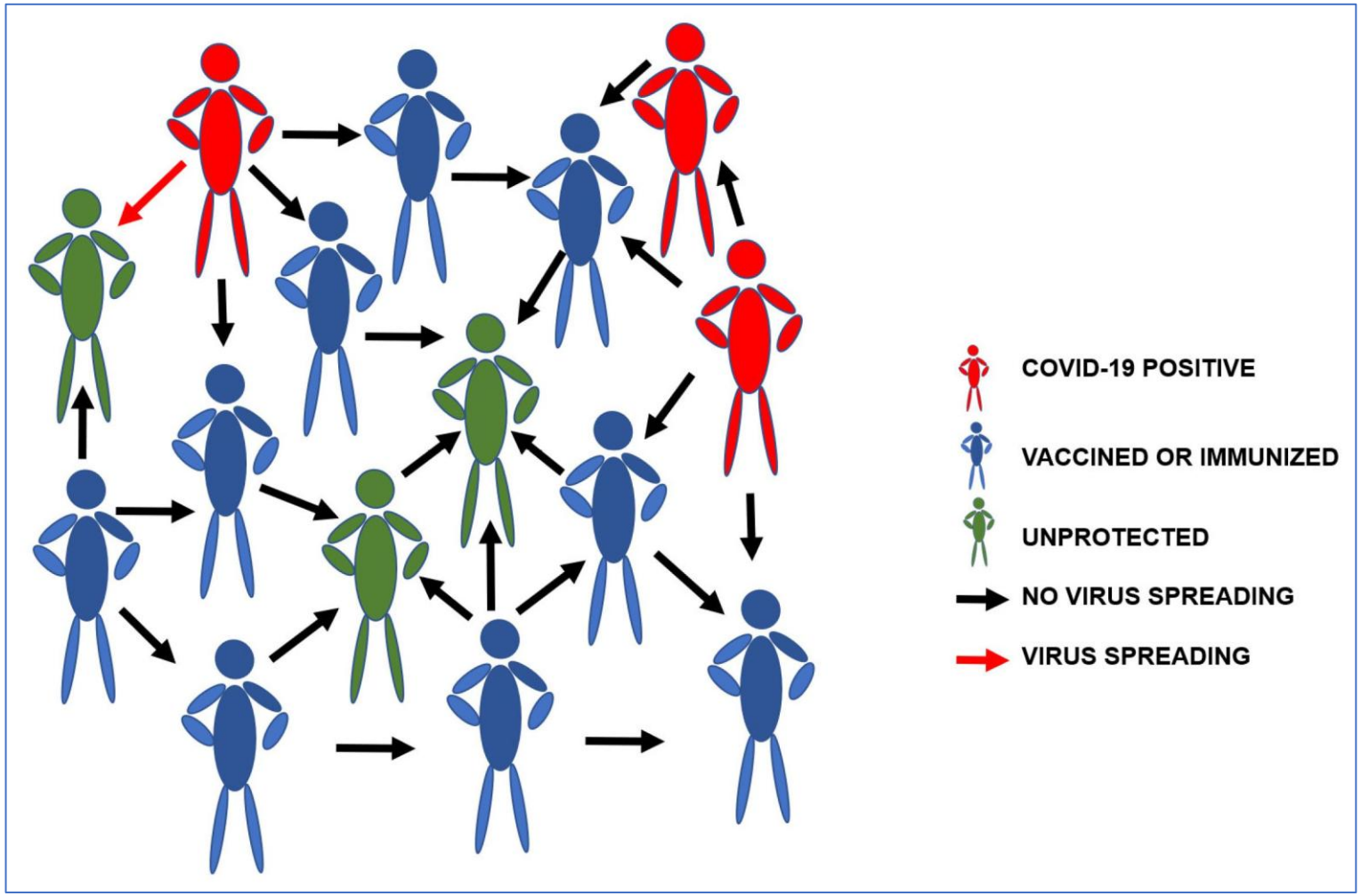
- **1. Problem Setting**
- **H**ow vaccination affects disease spread in a **contact network**.
- Let:
- $G = (V, E)$ →contact network
- $T_i \in \{0,1\}$ →vaccination status
- $Y_i \in \{0,1\}$ →infection outcome

Cont.

- **2. Why standard causal inference fails**
- In classical models:
- $Y_i = f(T_i)$
- In networks:
- $Y_i = f\left(T_i, ' \quad T_{\mathcal{N}(i)}\right)$
- E.g. Infection risk depends on:
- Your vaccination (**direct effect**)
- Your neighbors' vaccination (**spillover effect**)

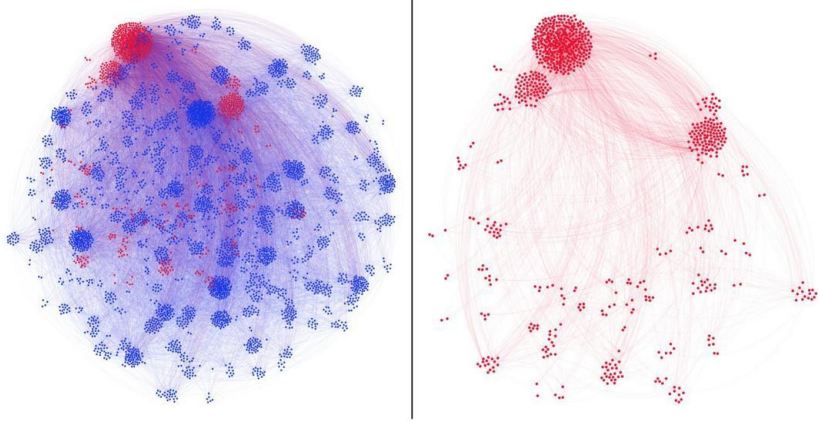
3. Visual intuition

- Vaccinated nodes act as barriers
- Infection paths get broken
- Unvaccinated nodes may still be protected → herd effect

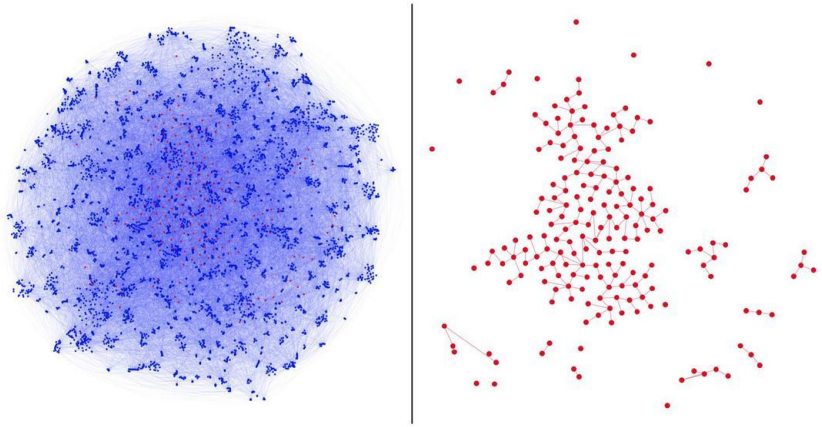


Cont.

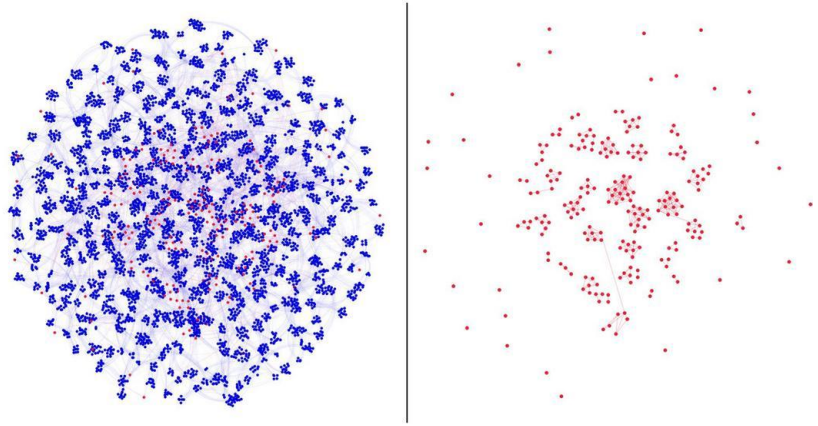
- Nodes → individuals
- Edges → contact/infection pathways
- Highlighted nodes → vaccinated individuals
- **observation:**
- Infection paths are **interrupted by vaccinated nodes**
- Some **unvaccinated nodes remain uninfected** because:
  - They are surrounded by vaccinated neighbors
- Vaccination not only provides **direct protection** to treated individuals but also generates **spillover protection** by disrupting transmission paths in the network



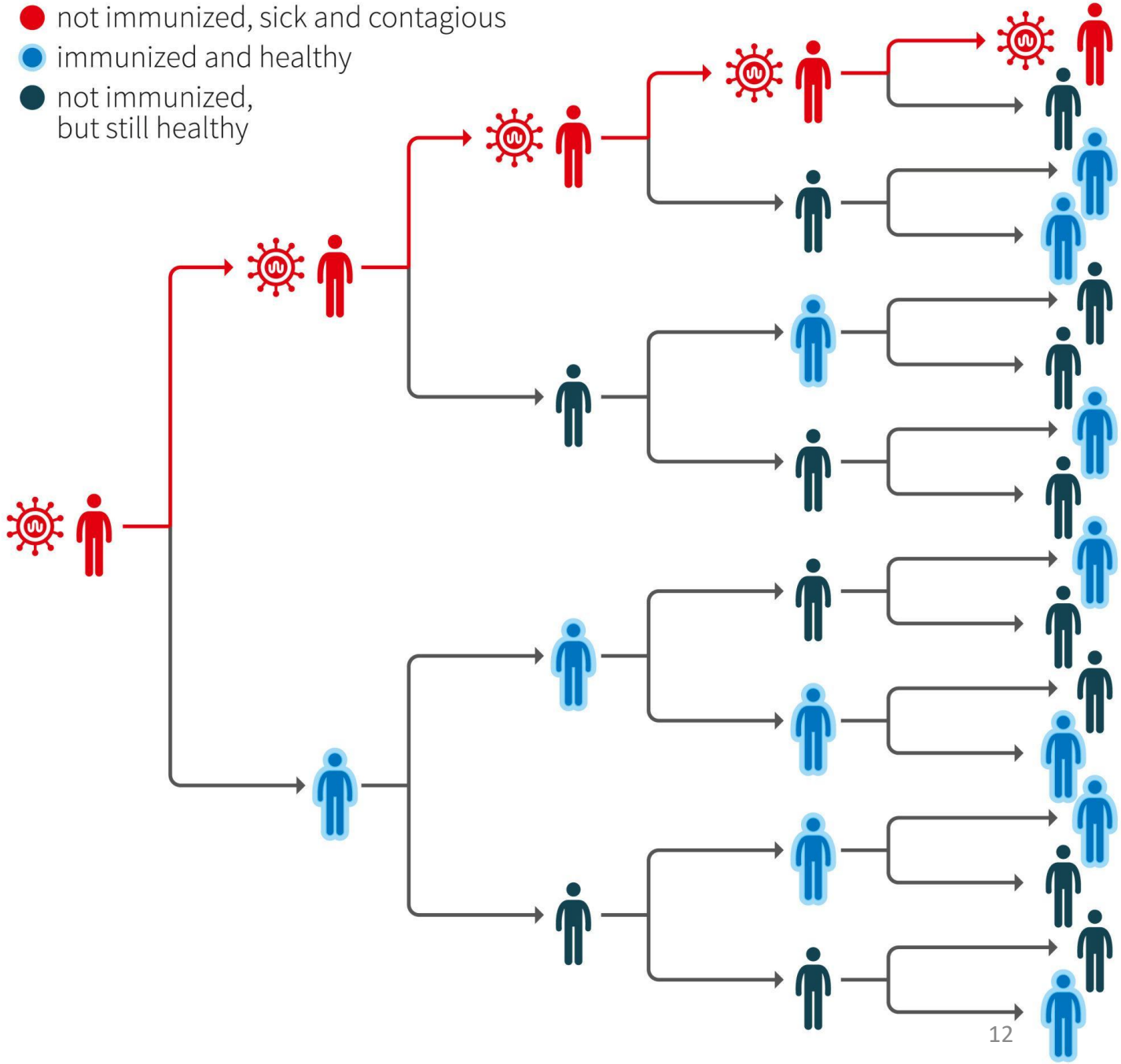
(A) SF network with one dominant anti-vaccine community.



(B) RN with one large and several smaller anti vaccine communities.



(C) SW with multiple small, disconnected anti-vaccine communities.



Interpretation

- **1. Infection without intervention (baseline logic)**
- Infection propagates along edges
- Any connected path can transmit disease
- If no one is vaccinated: Infection spreads through the largest connected component
- **2. Vaccinated nodes as "barriers"**
- Vaccinated nodes **block transmission paths**
- They act like **cut-points** in the graph
- A vaccinated node removes or weakens edges in the transmission graph
- **3. Local protection (Direct Effect)**
- If node i is vaccinated:
- Probability of infection reduces
- This is **individual-level causality**

Cont.

- **4. Spillover protection (Indirect Effect)**
- Now focus on an **unvaccinated node** surrounded by vaccinated neighbors.
- Infection paths **cannot reach it easily**
- Even without treatment, node i benefits from neighbors' treatment
- **5. Fragmentation of the network**
- Vaccination causes: The network to **break into smaller components**
- Large connected cluster → fragmented subgraphs
- This dramatically reduces epidemic spread
- **6. Herd immunity (global effect)**
- When enough nodes are vaccinated: The network no longer supports large-scale propagation
- Infection dies out quickly and Spread is **localized**

Cont.

- **Formal Interpretation**

$$Y_i = f(T_i, T_{\mathcal{N}(i)}, G)$$

Where:

- T_i: own vaccination
- $T_{\mathcal{N}(i)}$: neighbors' vaccination
- G: network structure

Cont.

- The figure illustrates how vaccination alters the effective connectivity of a contact network.
- Vaccinated individuals act as barriers that interrupt transmission paths, thereby reducing both direct infection risk and indirect risk through neighbors.
- As vaccination coverage increases, the network fragments, limiting large-scale spread and producing herd immunity effects.
- This demonstrates that causal effects in networks include both individual and spillover components.

4. Types of causal effects

Effect	Interpretation
Direct	Does vaccination protect the individual?
Indirect (spillover)	Does others' vaccination protect me?
Total	Combined protection
Network effect	Global reduction in spread

Cont.

- **5. Experimental Design**
- **Cluster Randomized Trial**
- Divide population into clusters (e.g., villages)
- Randomize vaccination at cluster level
- **Problem:**
- Infection spreads **across cluster boundaries**
- Violates independence → biased estimates

6. Exposure Mapping Approach

Define exposure:

$$E_i = \frac{\sum_{j \in \mathcal{N}(i)} T_j}{|\mathcal{N}(i)|}$$

Interpretation:

- Fraction of vaccinated neighbors

Now estimate:

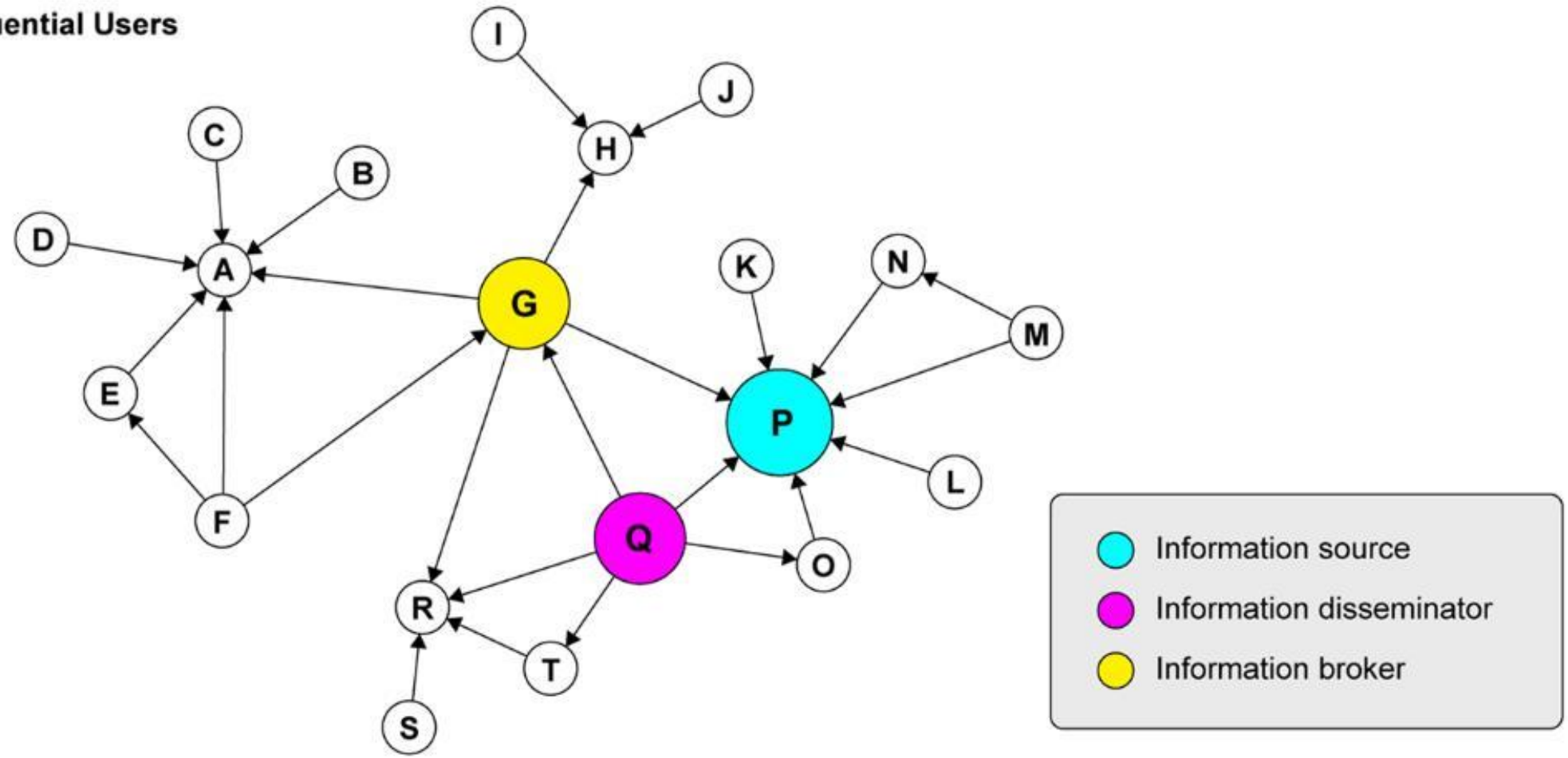
$$Y_i = f(T_i, E_i)$$

Cont.

- Empirical Insight - Indirect effects can be as large as or larger than direct effects
- Meaning: Vaccinating key nodes → protects entire network
- Policy Implication –

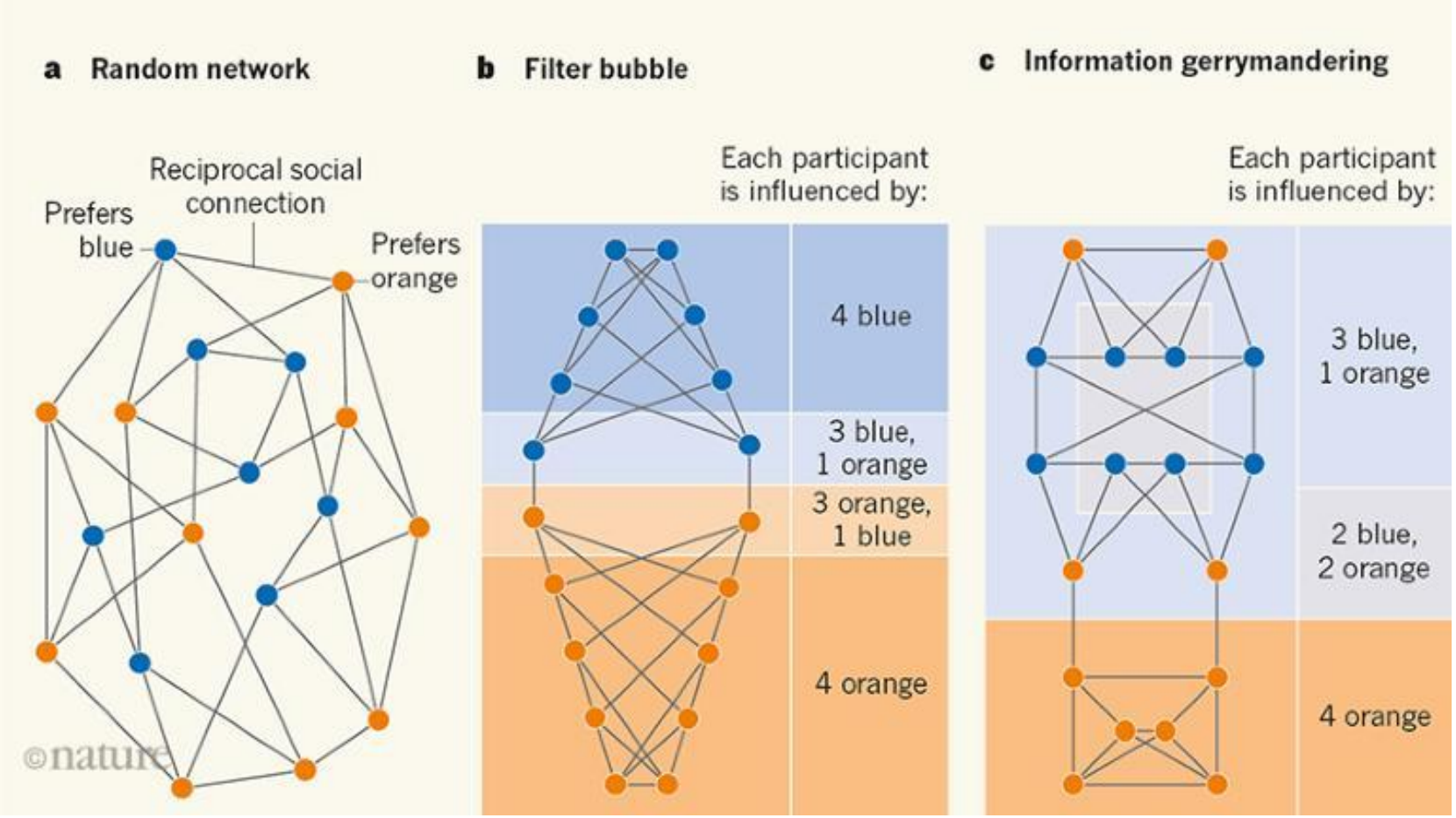
Strategy	Outcome
Random vaccination	Moderate impact
Targeted vaccination (high-degree nodes)	Strong impact

C. Influential Users



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
IN-DEGREE	6	0	0	0	1	0	2	3	0	0	0	0	1	0	1	**7**	0	4	0	1
OUT-DEGREE	0	1	1	1	1	3	4	0	1	1	1	1	1	2	1	0	**5**	0	1	1
BETWEENNESS	0	0	0	0	0	0	**5**	0	0	0	0	0	0	0	0	0	0	0	0	0

Cont.


a Random network
Reciprocal social connection
Prefers blue
Prefers orange
©nature
b Filter bubble
Each participant is influenced by:
4 blue
3 blue, 1 orange
3 orange, 1 blue
4 orange
c Information gerrymandering
Each participant is influenced by:
3 blue, 1 orange
2 blue, 2 orange
4 orange

Gerrymandering (meaning)

- Gerrymandering is the deliberate manipulation of electoral district boundaries to give one political party, group, or incumbent an unfair advantage.
- By "packing" opposition voters into a few districts or "cracking" them across many, the controlling party secures more seats than their vote share would typically warrant.
- **Origin:** The term originated in 1812 from Massachusetts Governor Elbridge **Gerry**, who approved a sala**mander**-shaped district.

Deep Link Entropy (DLE) for Edge Significance

- **Precise definition**
- **Deep Link Entropy** is an information-theoretic metric used on graphs (web graphs, knowledge graphs, identity graphs) to quantify the **informativeness or discriminative power of an edge**. It measures how **selective vs. generic** a link is.
- **Intuition**
- Not all edges carry equal meaning.
- “follows”, “likes”, “visited” → **high-frequency, low information**
- “signed this contract”, “owns this wallet”, “issued this credential” → **low-frequency, high information**
- DLE formalizes this using entropy. DLE acts as a **weighting function for edges in graph inference systems**.

More examples:

- **Identity graph**
- Edge: “has email”
  → appears everywhere → **high entropy → weak signal**
- Edge: “controls private key of wallet X”
  → rare, unique → **low entropy → strong signal**
- **Cybersecurity relevance**
- **1. Identity linking**
- Helps decide which signals to trust when linking identities
- **2. Fraud detection**
- Rare behavioural edges carry stronger evidence
- **3. Graph-based anomaly detection**
- Low-entropy edges → high suspicion or high confidence (context-dependent)

FONDUE Framework for Node Disambiguation / Deduplication

- **Definition**
- **FONDUE** (commonly expanded as ***Framework for ONline Data Unification and Entity resolution***) is a **graph-based entity resolution framework** used to:
- Identify when **multiple nodes refer to the same real-world entity**
- Merge or link them appropriately
- **E.g.**
- In real systems: Same user appears as:
  - Email ID
  - Wallet address
  - Device fingerprint
  - Username
- These are **distinct nodes but may represent the same entity**

Cont.

- **What FONDUE does -** It resolves **node ambiguity** using **multi-signal graph reasoning**
- **Key components**

1. Feature extraction **Node attributes:** **Email** **IP** **Device ID** **Behavioral patterns**	**2. Edge construction** **Build relationships:** **"same IP"** **"same device"** **"transaction link"**	**3. Edge weighting (where DLE fits)** **Assign importance using:** **Deep Link Entropy** **Frequency** **Reliability**
4. Similarity scoring Compute: Similarity(u, v) $= f(\textit{shared edges}, \textit{attributes}, \textit{weights})$		**5. Clustering / merging** If similarity > threshold: Merge nodes Or create "same-as" links

Cybersecurity applications

1. Fraud & Sybil detection

- Detect fake accounts controlled by same actor

2. **Blockchain analytics**

- Cluster wallet addresses into entities

3. Threat intelligence

- Link attacker infrastructure:
  - Domains
  - IPs
  - Malware signatures

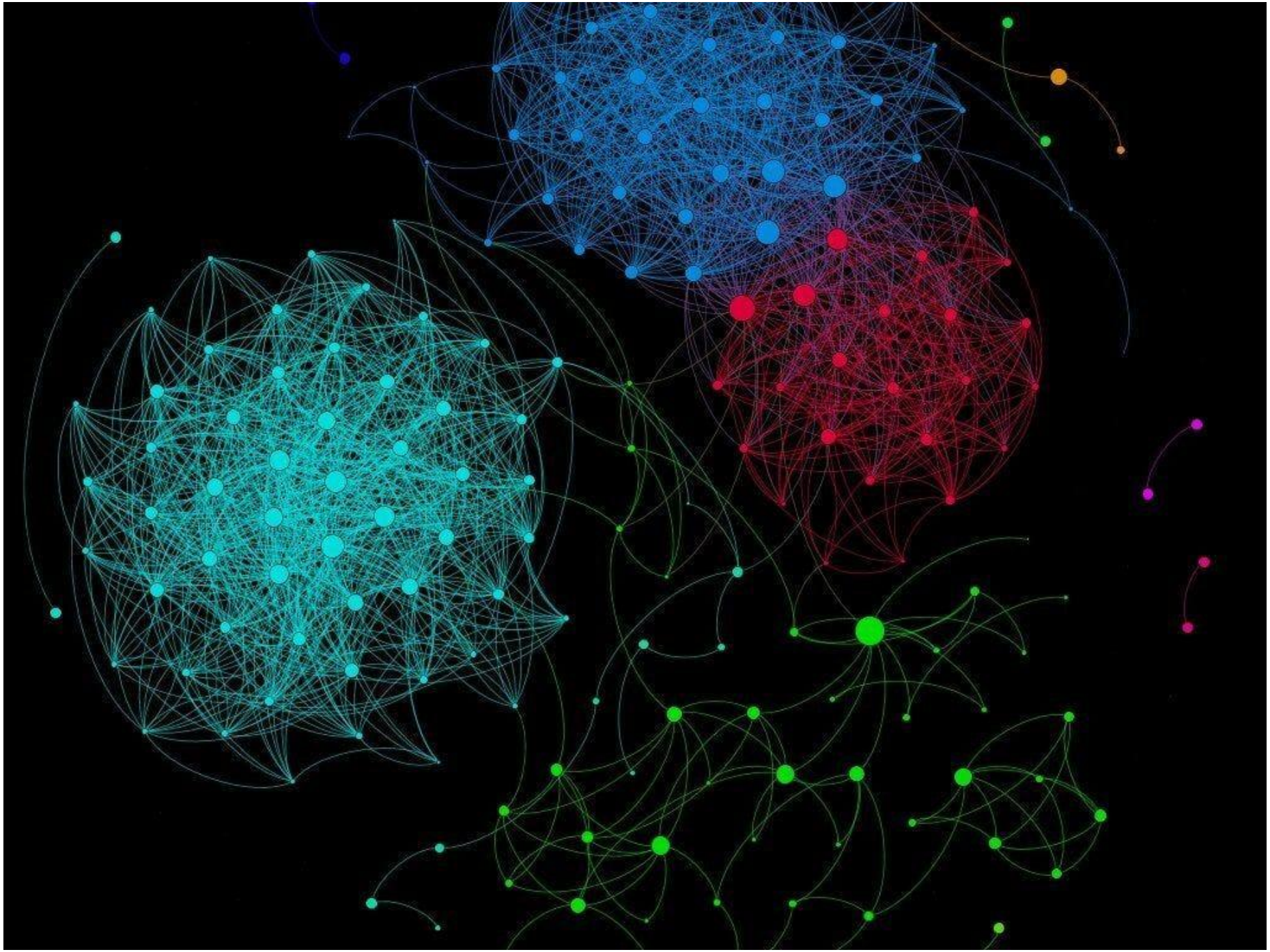
Summary

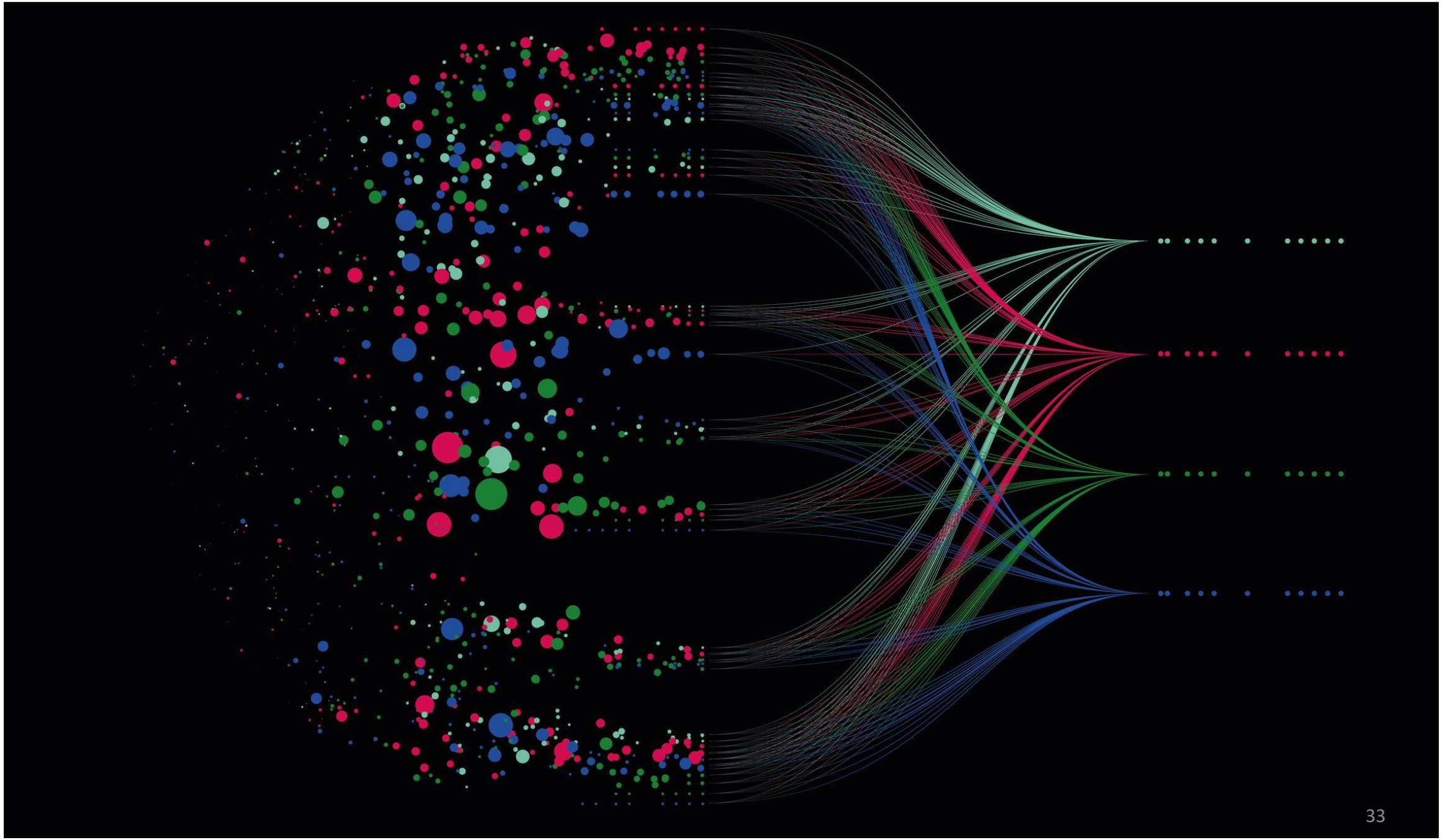
Concept	Role
Deep Link Entropy	Measures **edge importance**
FONDUE	Uses edges to **merge nodes (entities)**

- Deep Link Entropy tells you *which connections matter*.
- FONDUE uses those connections to decide *which identities are actually the same*.
- Together, they form a powerful foundation for:
- Identity resolution
- Fraud detection
- Trust graph construction

Collaborative Dynamics & Cultural Networks

- Real-world systems are networked, not isolated
- Key questions:
- How do collaborations form?
- How does knowledge evolve collectively?
- Tools:
- Statistical network models
- Large-scale empirical datasets (e.g., Wikipedia)

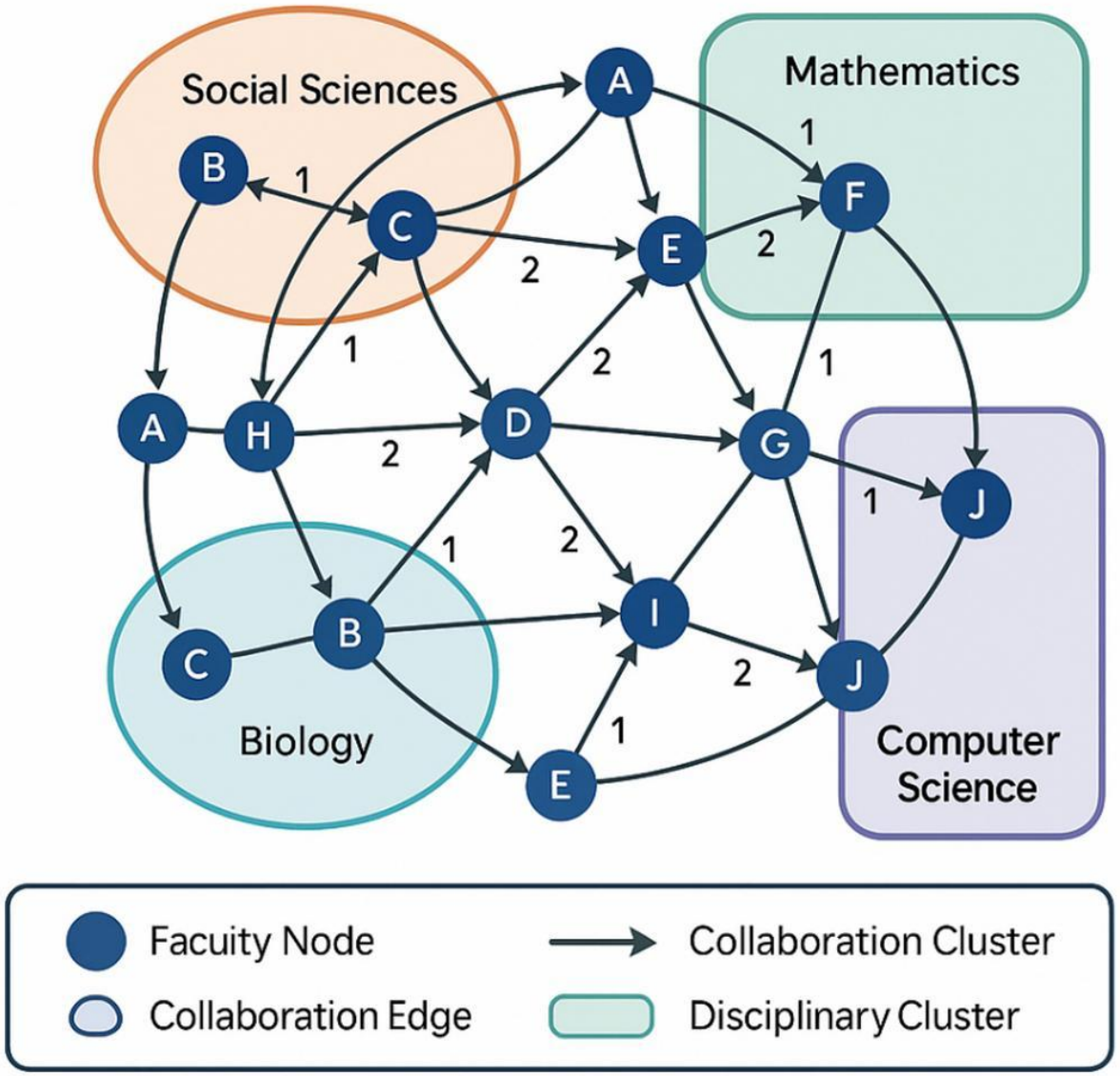




Application 1: Collaborative Networks

- **Research Partnerships as Networks**
- Nodes → Researchers
- Edges → Co-authorship / collaboration
- Observations:
  - Clustering (research groups)
  - Preferential attachment (elite researchers attract more links)
  - Homophily (same field, institution)

See next figure..


Social Sciences
Mathematics
Biology
Computer Science
A
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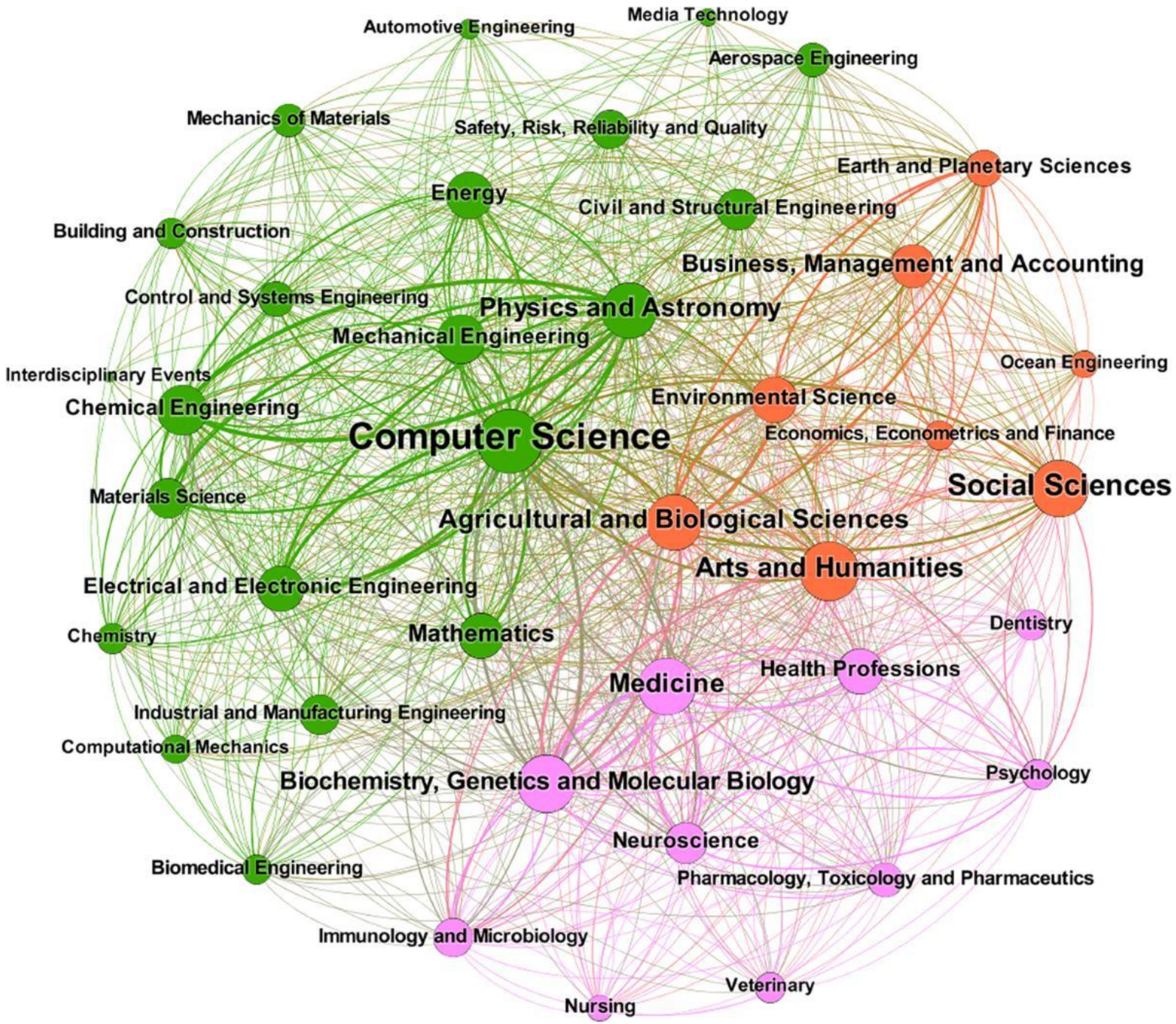
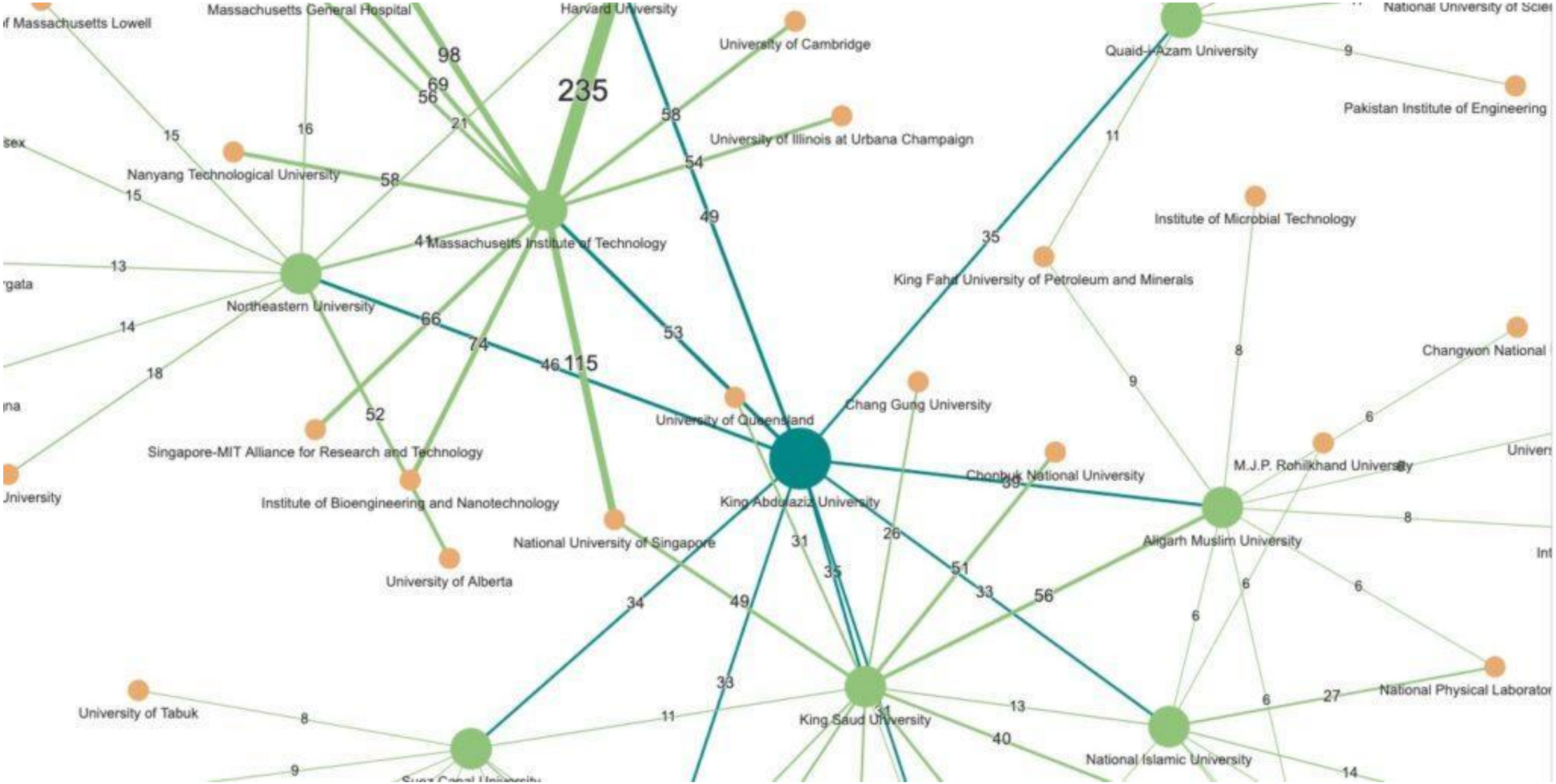
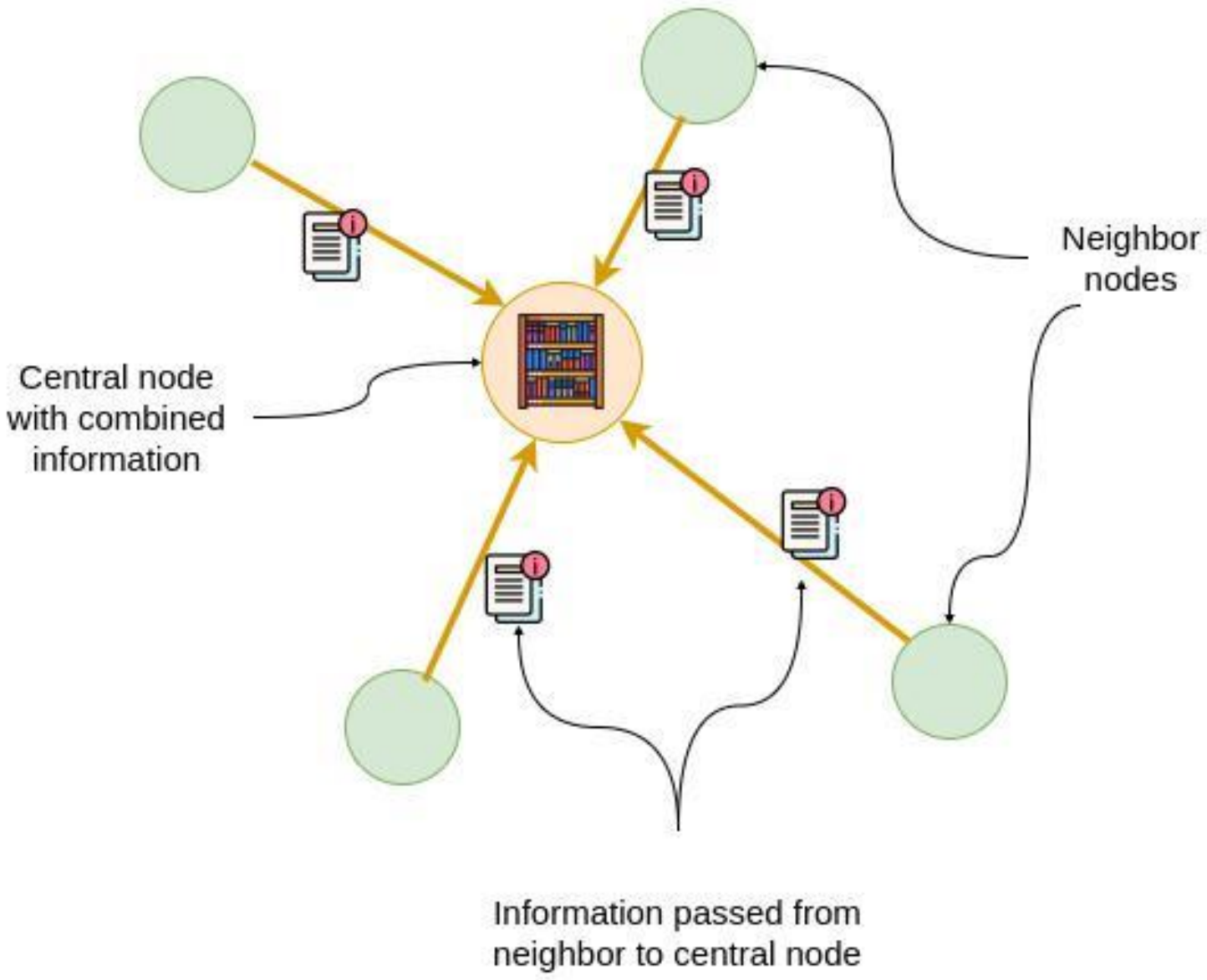
Facuity Node
Collaboration Edge
Collaboration Cluster
Disciplinary Cluster


Automotive Engineering
Media Technology
Aerospace Engineering
Mechanics of Materials
Safety, Risk, Reliability and Quality
Earth and Planetary Sciences
Energy
Civil and Structural Engineering
Building and Construction
Business, Management and Accounting
Control and Systems Engineering
Physics and Astronomy
Mechanical Engineering
Ocean Engineering
Interdisciplinary Events
Environmental Science
Chemical Engineering
Computer Science
Economics, Econometrics and Finance
Social Sciences
Materials Science
Agricultural and Biological Sciences
Arts and Humanities
Electrical and Electronic Engineering
Dentistry
Chemistry
Mathematics
Health Professions
Medicine
Industrial and Manufacturing Engineering
Computational Mechanics
Psychology
Biochemistry, Genetics and Molecular Biology
Neuroscience
Biomedical Engineering
Pharmacology, Toxicology and Pharmaceutics
Immunology and Microbiology
Veterinary
Nursing


Massachusetts General Hospital
Harvard University
University of Cambridge
Quaid-i-Azam University
Pakistan Institute of Engineering
University of Illinois at Urbana Champaign
Nanyang Technological University
Massachusetts Institute of Technology
Institute of Microbial Technology
King Fahd University of Petroleum and Minerals
Northeastern University
Changwon National
Chang Gung University
University of Queensland
Singapore-MIT Alliance for Research and Technology
Chonbuk National University
M.J.P. Rohilkhand University
Institute of Bioengineering and Nanotechnology
King Abdulaziz University
National University of Singapore
Aligarh Muslim University
University of Alberta
National Physical Laboratory
University of Tabuk
King Saud University
National Islamic University
235
115

Application: Wikipedia as a Cultural Network

- **Network Representation:**
- Nodes → Articles
- Edges → Hyperlinks / references
- **Alternative views:**
- Editor collaboration network
- Topic similarity network

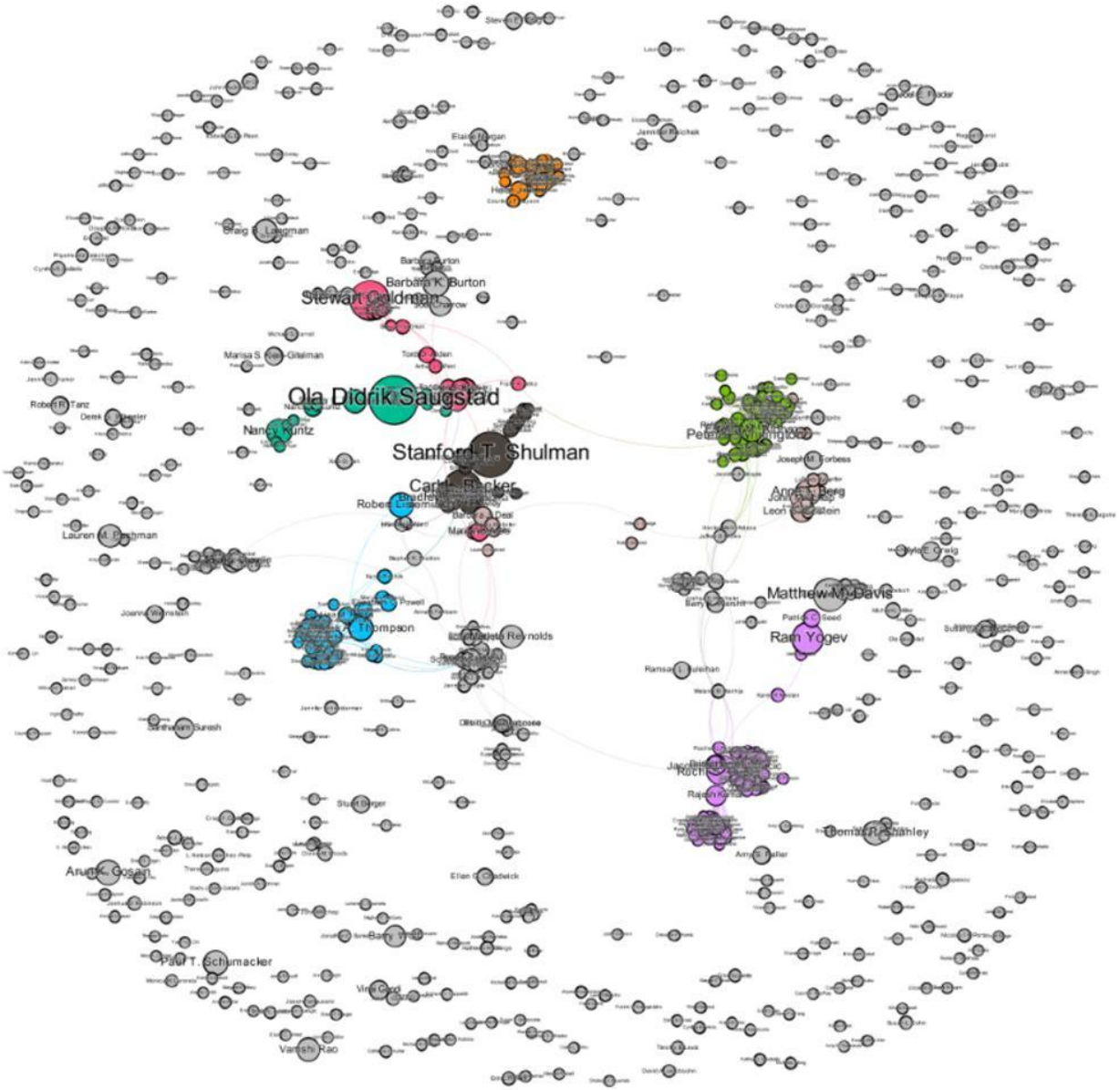

Neighbor nodes
Central node with combined information
Information passed from neighbor to central node

What Makes Wikipedia Unique

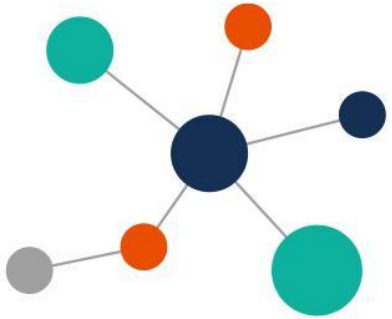
- Decentralized content creation
- Continuous evolution
- Multi-language structure
- Acts as:
- A **living cultural graph of human knowledge**

Network Properties of Wikipedia

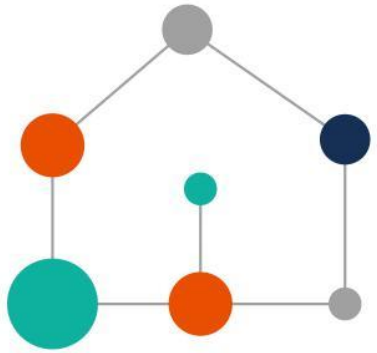
- Highly **connected core topics**
- Strong **hierarchical structure**
- **Power-law degree distribution**
- Emergence of:
  - Authority pages
  - Knowledge hubs



Co-authorship network types



Centralized

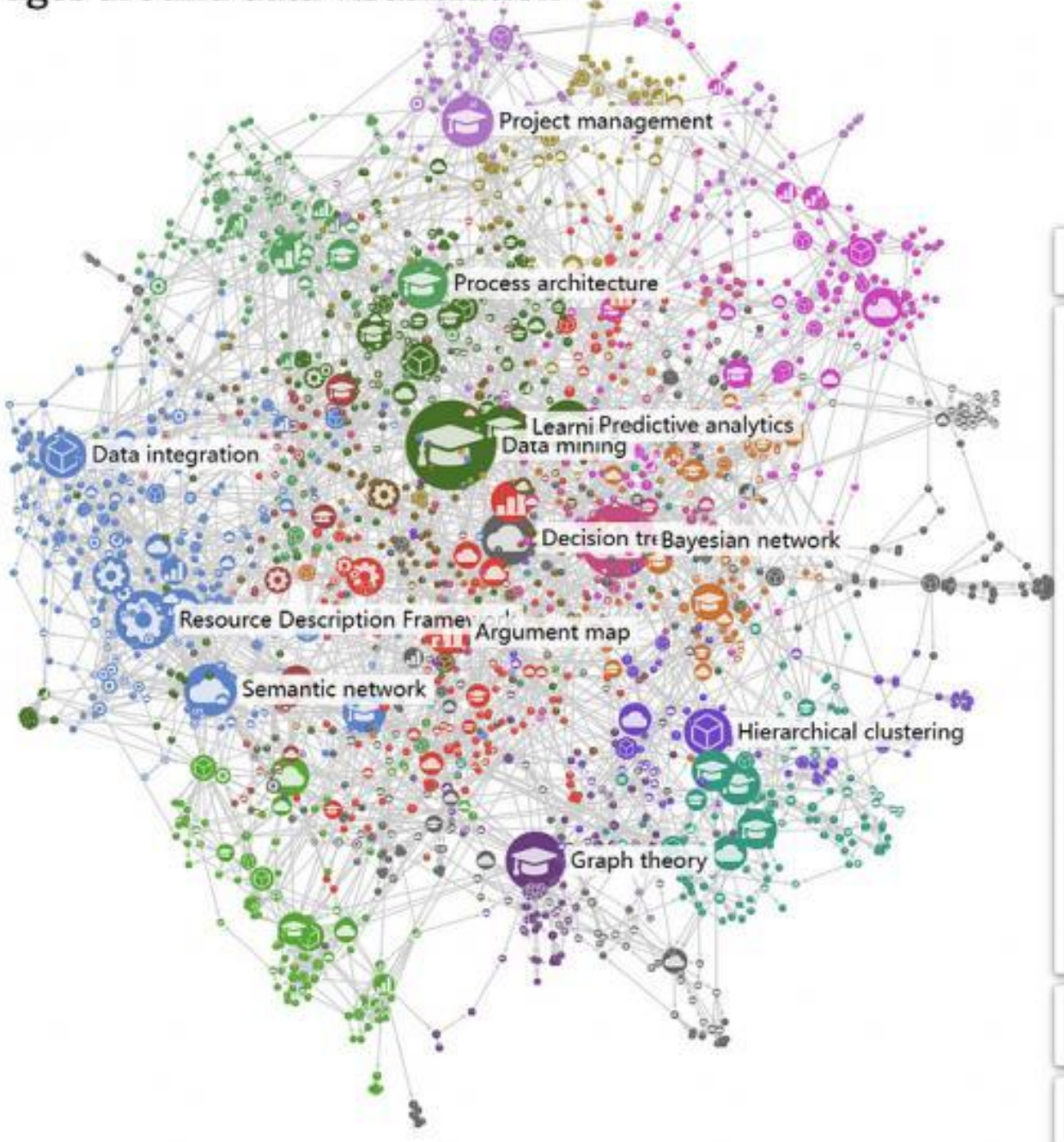


Sparse

Fully-connected

A cartography of Wikipedia pages around data visualization

2085 nodes , 5409 edges



Search in nodes...

Description

This map represents a *network* of Wikipedia articles around the topic of "Data vizualisation". Each *node* represents an article, and each edge a "See also" link.

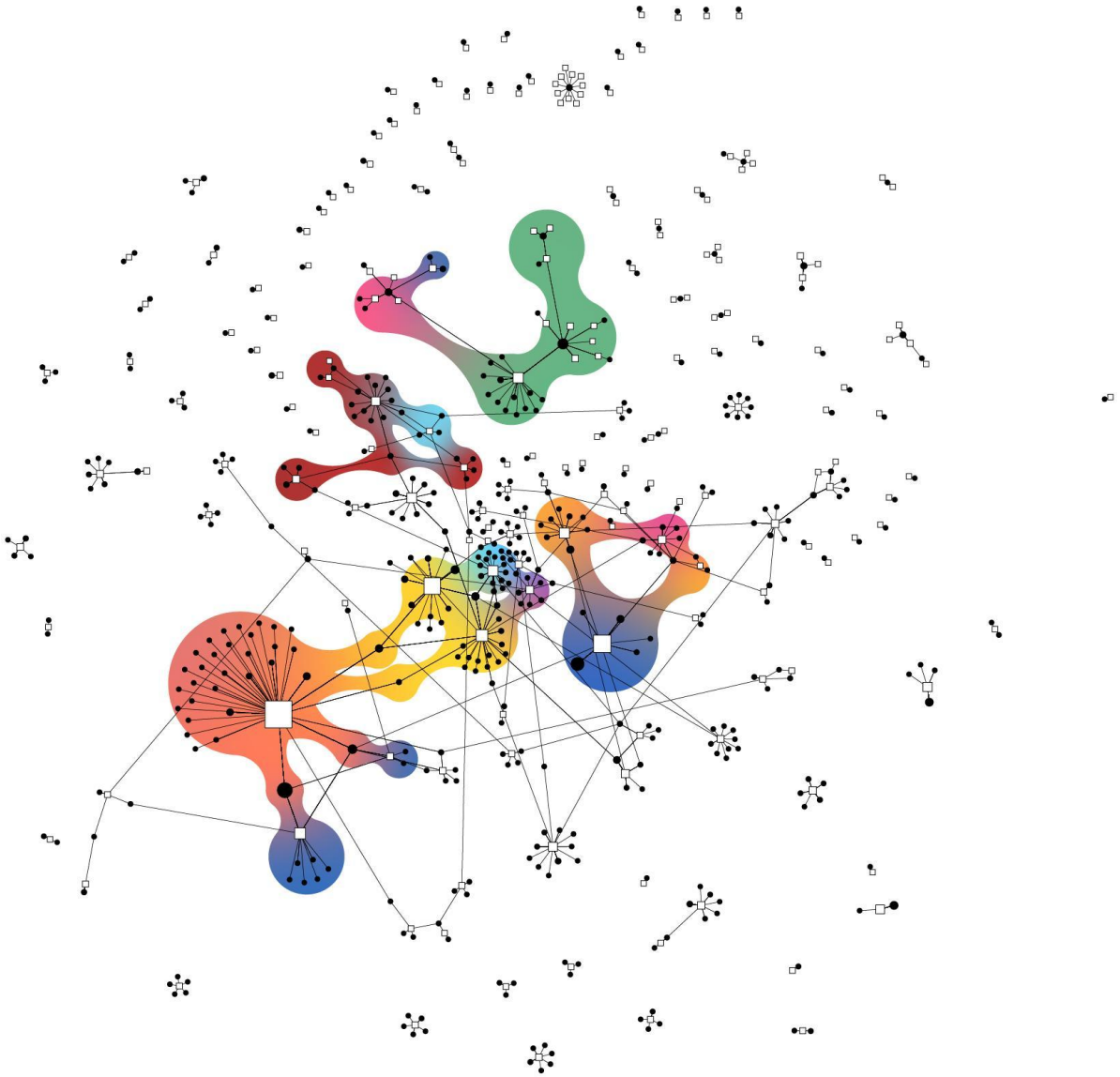
The seed articles were selected by hand by the Sciences-Po médialab team, and the network was crawled using Seealsology, and then cleaned and enriched manually. This makes the dataset creditable to both the médialab team and Wikipedia editors.

This web application has been developed by QuestWare, using react and sigma.js. You can read the source code on GitHub.

Nodes sizes are related to their betweenness centrality. More central nodes (ie. bigger nodes) are important crossing points in the network. Finally, You can click a node to open the related Wikipedia article.

Clusters

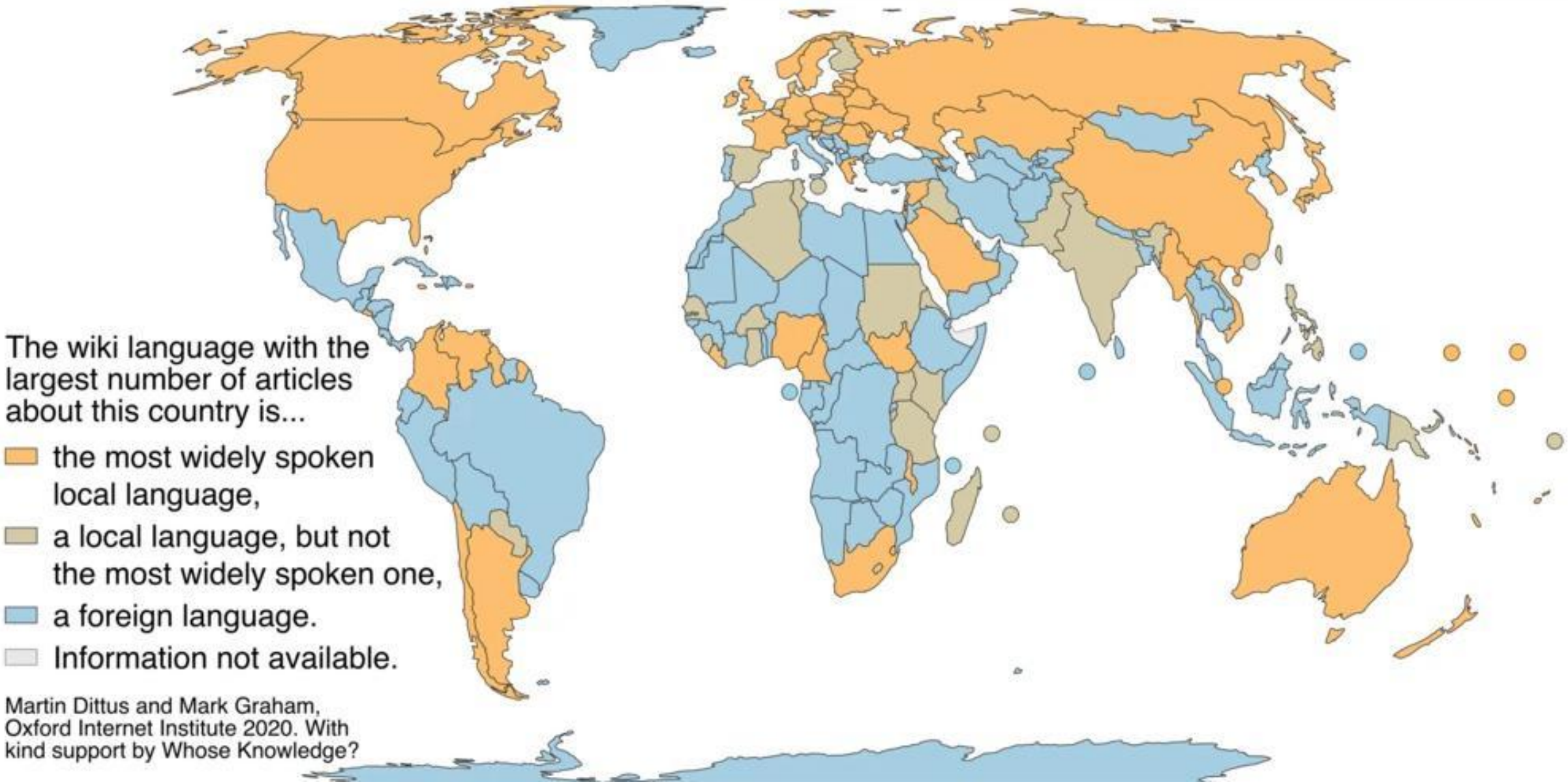
Categories



Donald Trump
United Kingdom
WWI
WWII
China
Barak Obama
France
Global Warming
United States

Cultural Insights

- Reflects:
  - Global knowledge priorities
  - Cultural biases
  - Language-specific structures
- Example:
  - Different language editions → different network topology


The wiki language with the largest number of articles about this country is...
the most widely spoken local language,
a local language, but not the most widely spoken one,
a foreign language.
Information not available.
Martin Dittus and Mark Graham, Oxford Internet Institute 2020. With kind support by Whose Knowledge?

Comparative Insight

Aspect	**Research Networks**	**Wikipedia**
Nature	Collaboration	Knowledge
Drivers	Social + institutional	Collective editing
Structure	Clusters, elites	Hubs, hierarchies
Dynamics	Slow evolution	Rapid updates

Cont.

- Wikipedia does not “guarantee truth” in a centralized sense; instead, it **operationalizes epistemic reliability through policy, community processes, and algorithmic support**.
- Think of it as a **self-correcting socio-technical system**.

Core Governance Model - Wikipedia

- At the platform level, Wikipedia enforces three foundational content policies:
- **(a) Neutral Point of View (NPOV)**
- Articles must represent **all significant viewpoints proportionally**
- Prevents ideological capture
- **(b) Verifiability**
- Claims must be **backed by reliable, published sources**
- Truth is proxied by **citability**, not personal belief
- **(c) No Original Research**
- Editors cannot introduce new theories
- Content must reflect **existing knowledge consensus**

Real time editing

- **Human Layer**
- Thousands of volunteer editors:
  - Watchlists (track changes on pages)
  - Patrol edits (especially new contributors)
- Domain experts often oversee specific topics
- **Automated Layer**
- Bots (e.g., ClueBot NG):
  - Detect vandalism
  - Revert malicious edits within seconds
- Monitoring is **distributed, persistent, and redundant**

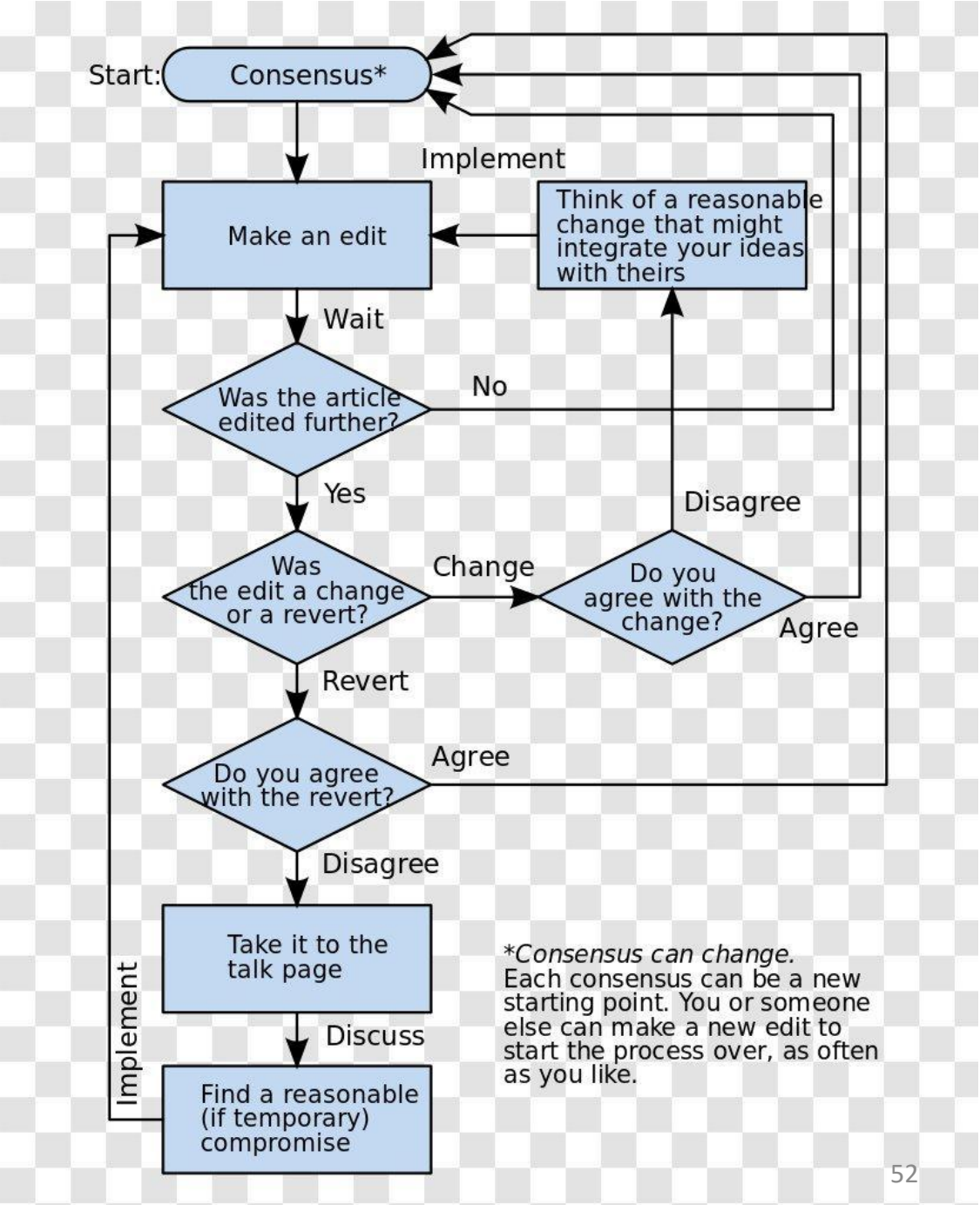
Reputation & Access Control

- Reputation = edit history + community trust

Role	Capability
Anonymous users	Can edit (with limits)
Registered users	Build edit history
Autoconfirmed users	Fewer restrictions
Administrators	Page protection, conflict resolution

Cont.

- Wikipedia resolves disputes via:
- Talk pages (discussion forums)
- Request for Comments (RfC)
- Editorial consensus
- **Quality Control Signals**
- **(a) Page Status Labels**
- Featured Article (highest quality)
- Good Article
- Stub / Needs improvement
- **(b) Citation density**
- More citations → higher reliability signal
- **(c) Edit stability**
- Stable pages → fewer reverts → higher trust



Cont.

- Wikipedia can be modeled as:
- **Editor network** → collaboration graph
- **Article network** → knowledge graph
- Key SNA insights:
- Central editors act as **hubs**
- Consensus emerges via **network interaction dynamics**

Takeaways

- Networks encode **hidden structure in human activity**
- ERGM reveals **mechanisms of link formation**
- Wikipedia shows:
- Knowledge is **networked, evolving, and culturally shaped**

ERGM - Exponential Random Graph Models

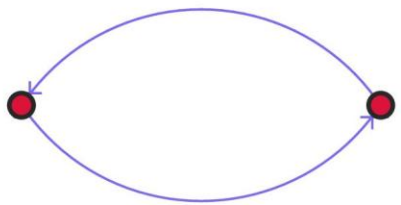
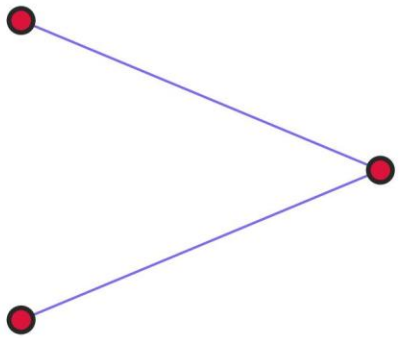
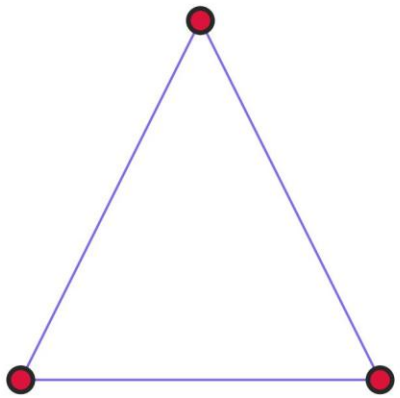
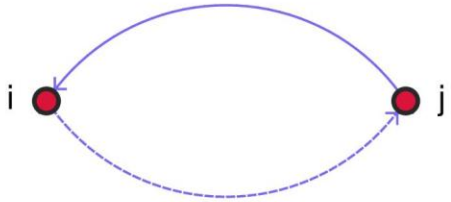
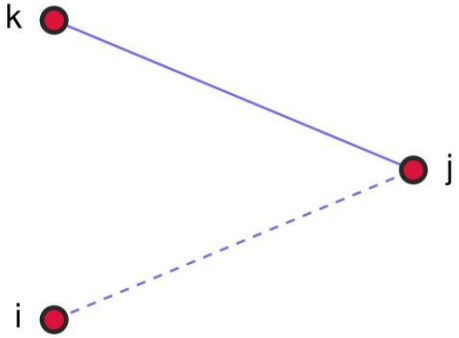
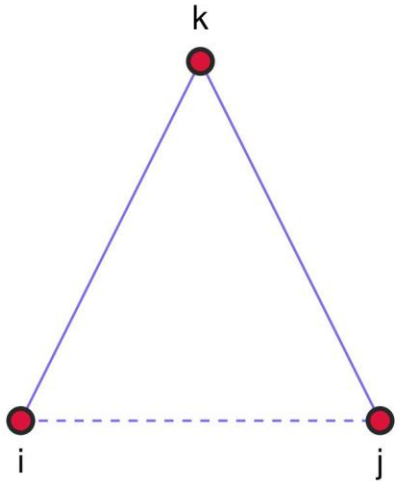
- Standard regression asks: given attributes of nodes, can we predict an outcome?
- ERGM asks something fundamentally different: given the *structure* of a network, what is the probability that *this particular network* exists among all possible networks on n nodes?
- The formula $P(G) \propto \exp(\theta^T s(G))$ is a Gibbs distribution
- the probability of observing network G depends exponentially on a set of network statistics s(G), weighted by parameters θ.
- The Parameters θ - Positive θ on triangles means triadic closure is more likely than chance. Negative θ on edges means the network is sparse. Zero means that feature plays no role.

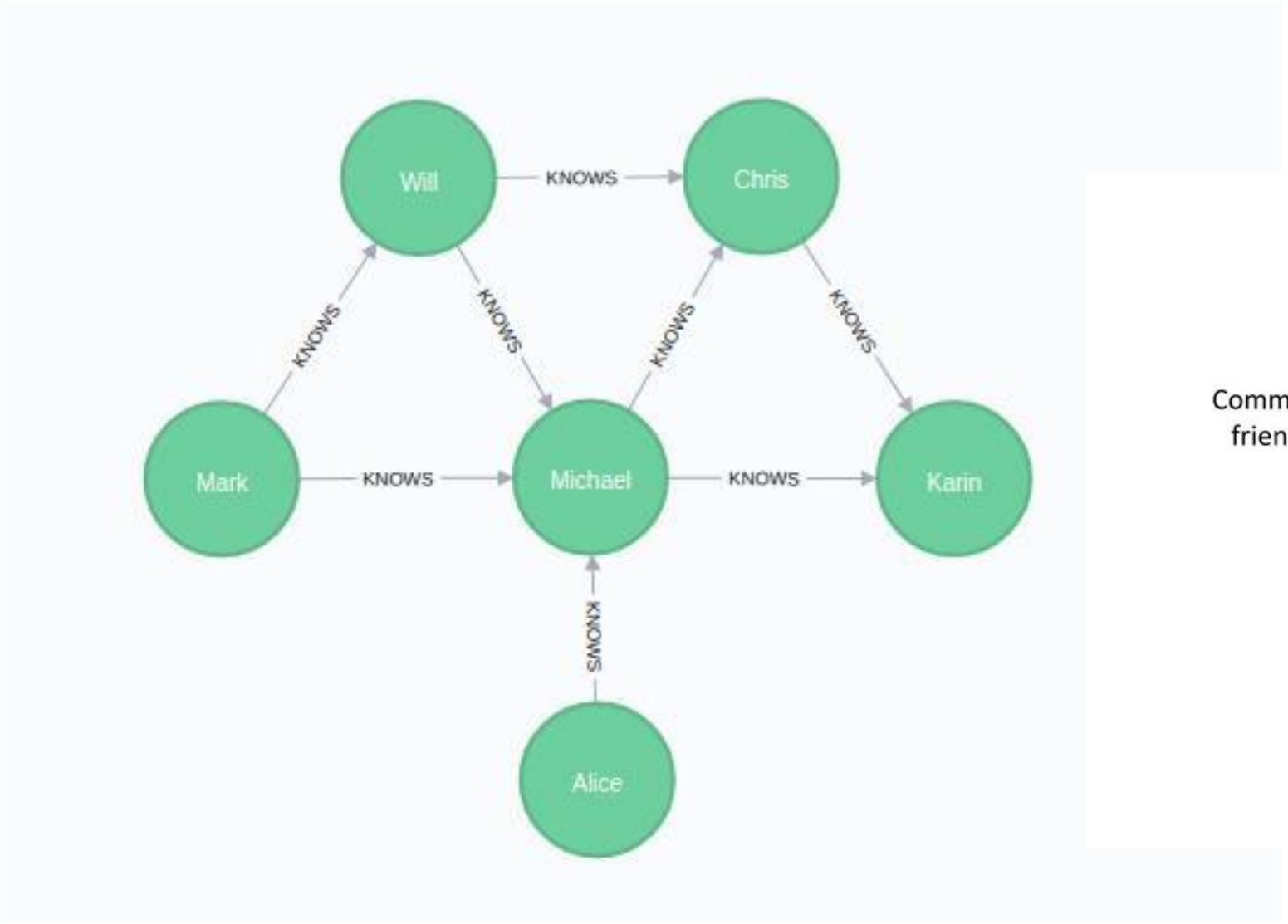
Exponential Random Graph Models (ERGM)

- $P(G) \propto \exp(\theta^T s(G))$
- Where:
- G: observed network
- $s(G)$:network statistics (edges, triangles, etc.)
- θ: parameters

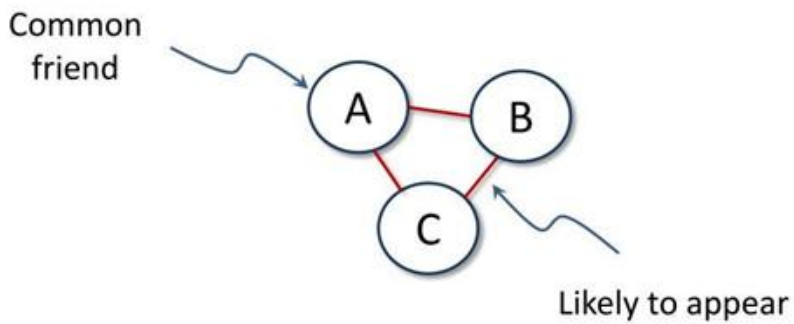
Exponential Random Graph Models (ERGM)

- **The Network Statistics s(G)**
- Typical statistics include
- edges (baseline density),
- triangles or triadic closure (do friends of friends become friends?),
- reciprocity in directed networks (if A → B, does B → A?),
- degree distribution terms, and
- node-level covariates like homophily (same department, same country).
- Each statistic captures a *structural tendency* you believe is driving link formation.

	Reciprocity	Preferential Attachment	Triadic Closure
Network			
Dyad$_{ij}$			



Triadic Closure

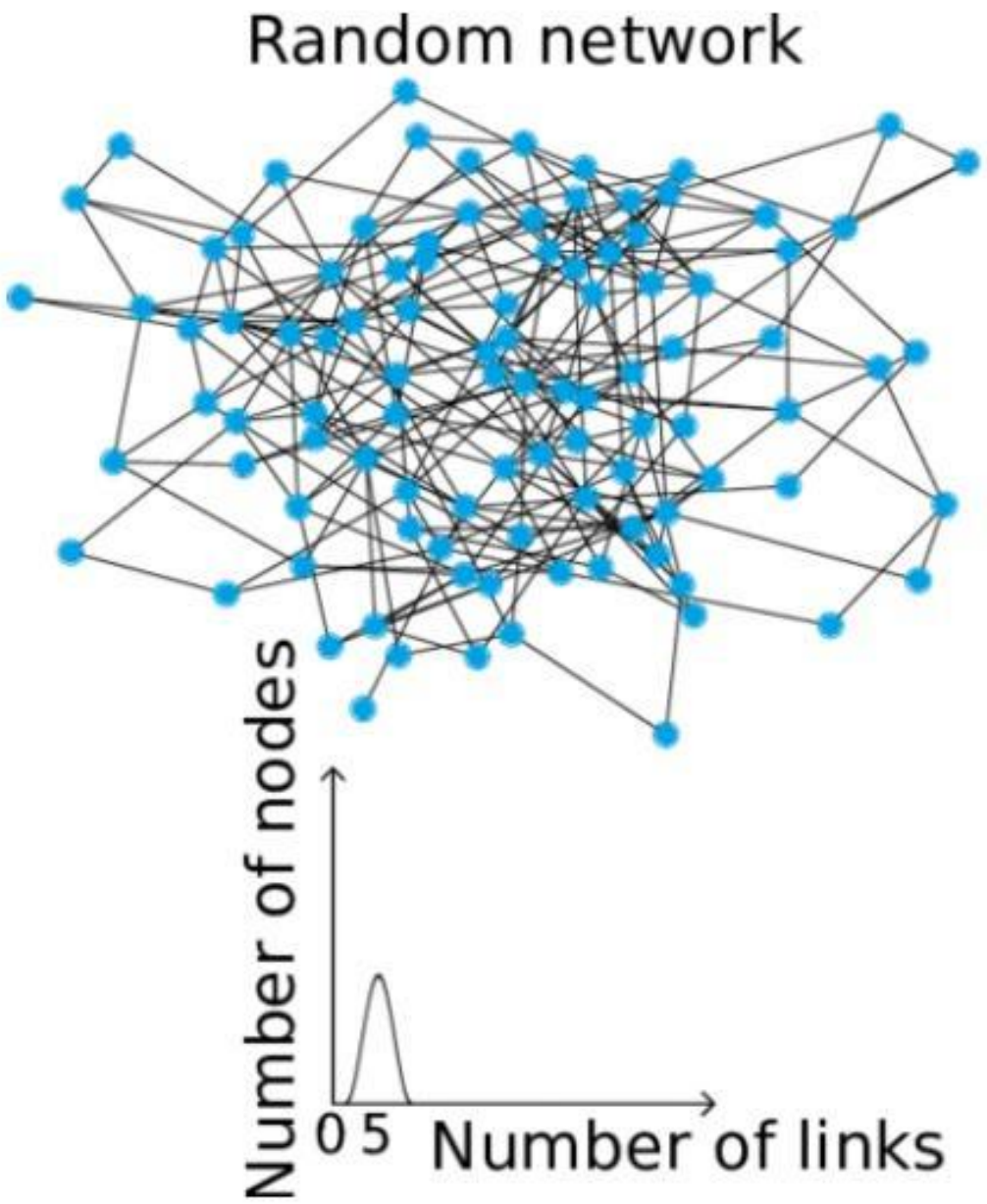


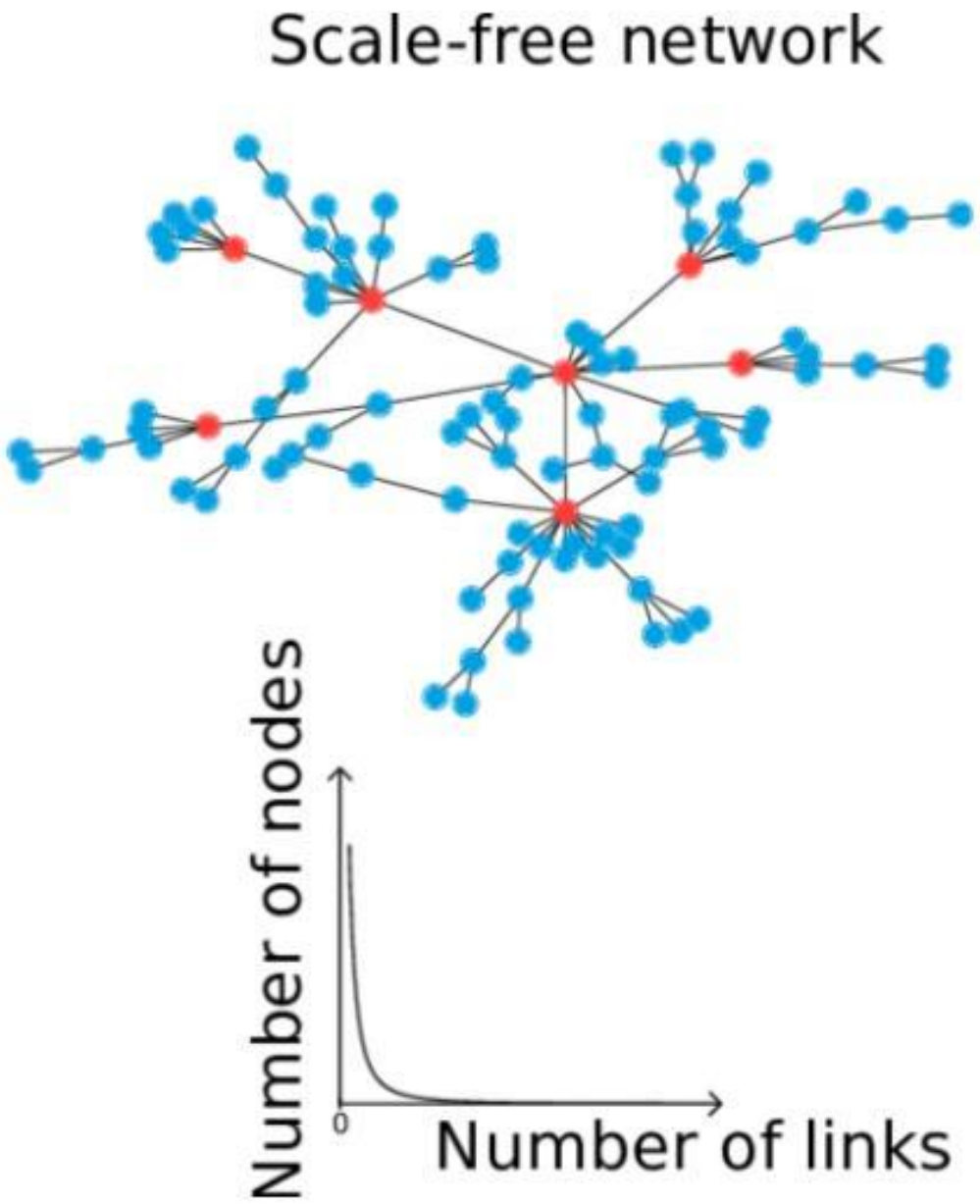
What ERGM Captures

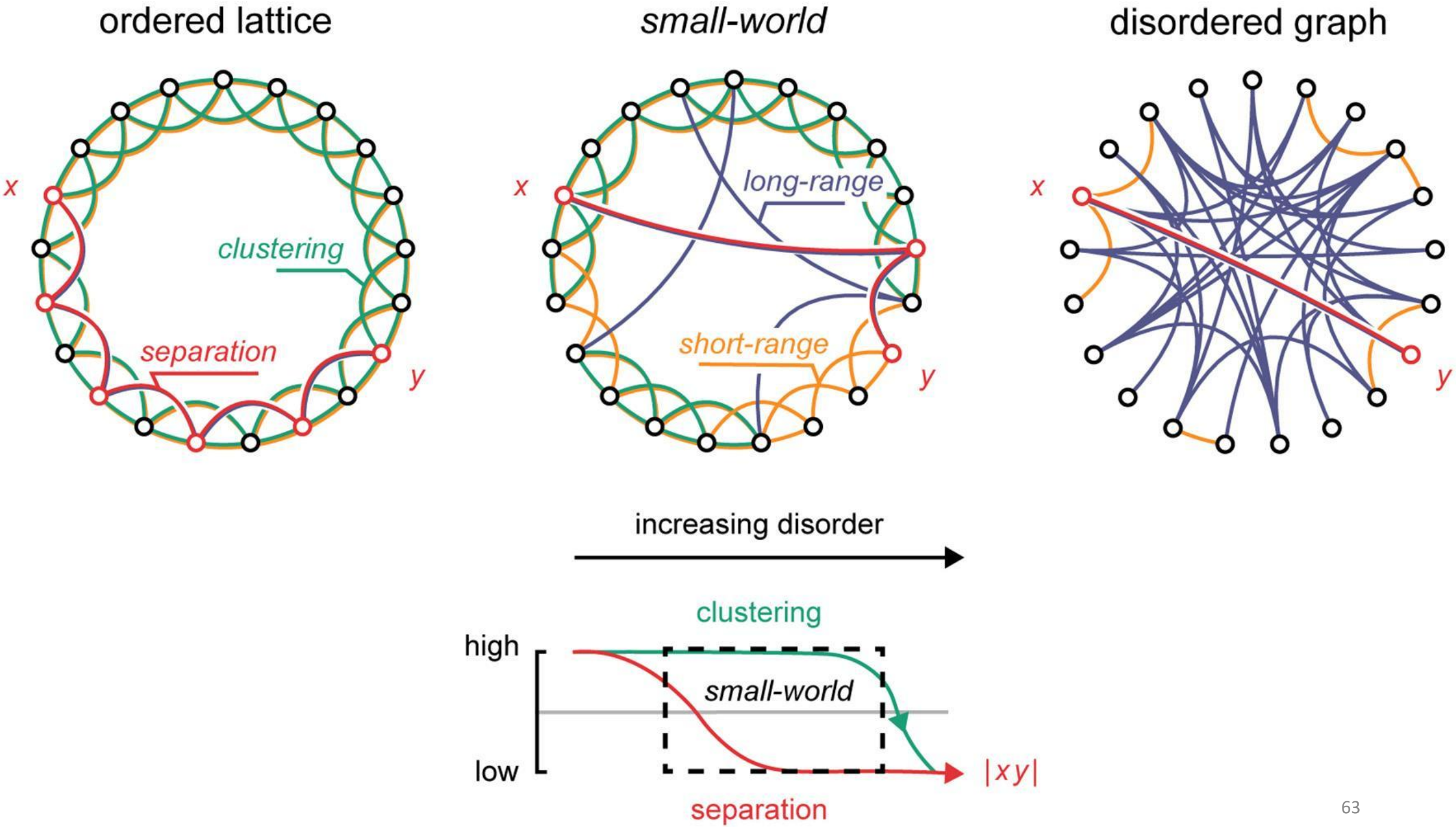
- **Edge formation probability**
- **Triadic closure** (collaboration triangles)
- **Reciprocity**
- **Attribute effects** (discipline, geography)
- Collaboration is **not random** — it is structurally driven

Findings in Research Networks

- Strong **triadic closure** → teams form clusters
- **Central actors dominate collaborations**
- Institutional proximity matters
- Collaboration networks show:
  - **Small-world properties**
  - **Scale-free behavior**

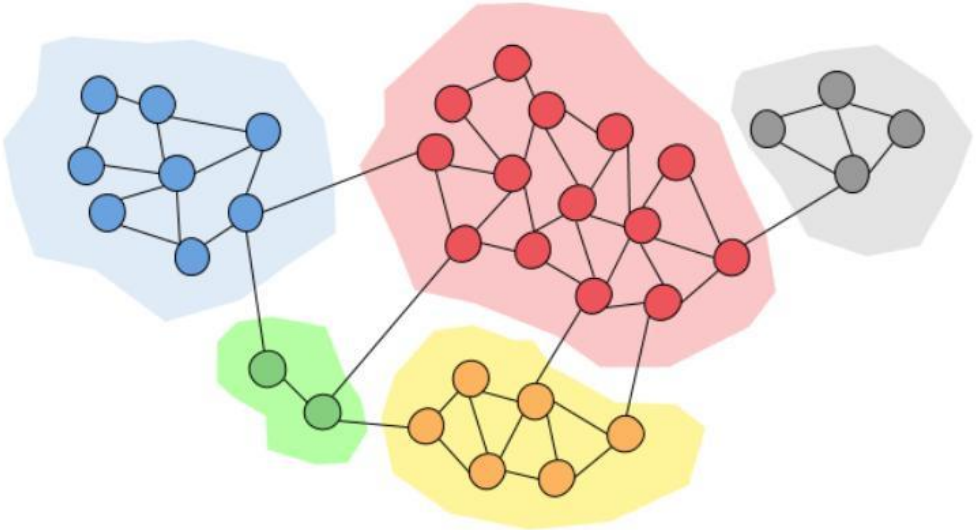

Random network
Number of nodes
0 5
Number of links

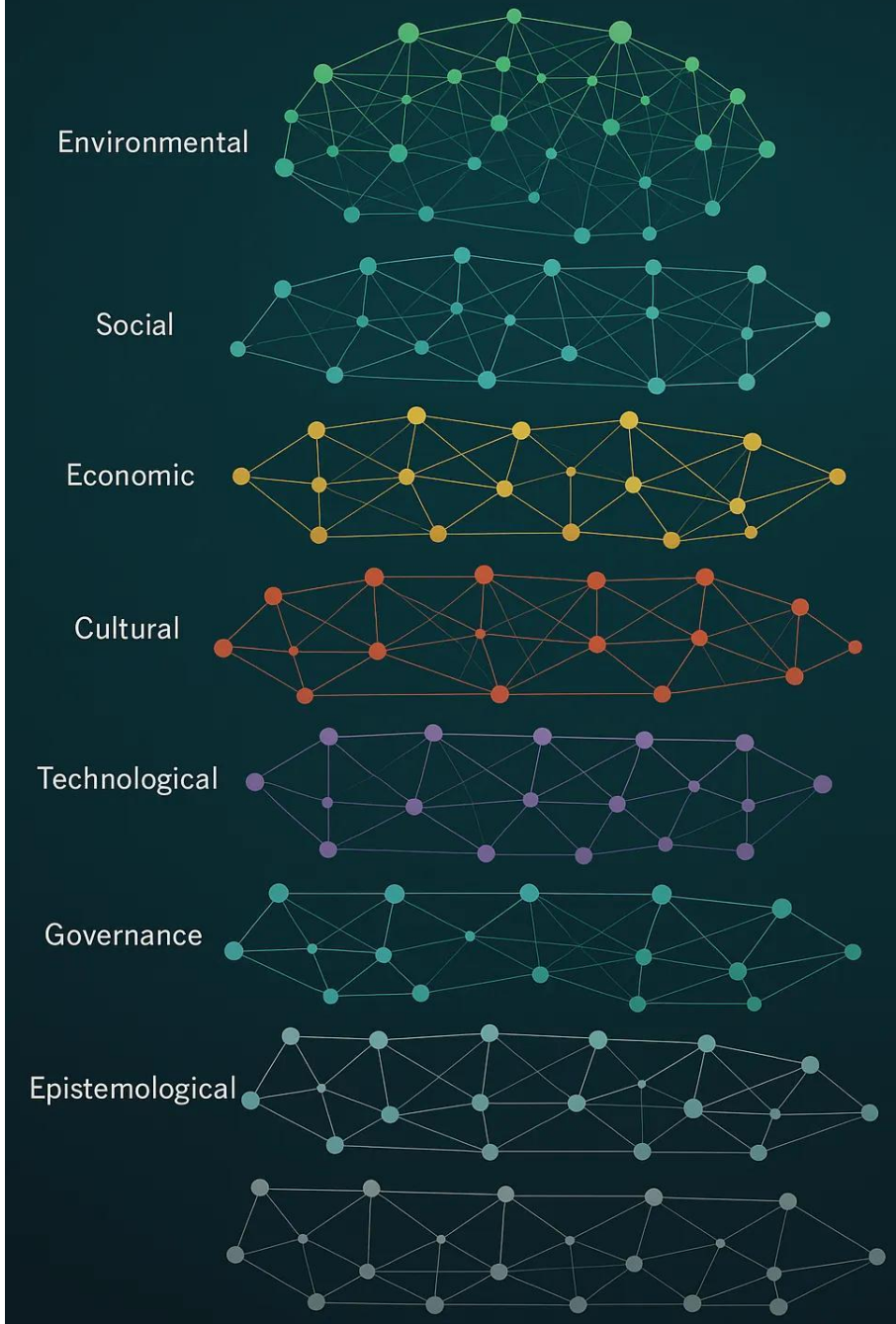

Scale-free network
Number of nodes
0
Number of links


ordered lattice
small-world
disordered graph
x
y
clustering
separation
long-range
short-range
increasing disorder
high
low
small-world
|xy|

Population-Scale Network Mapping

- **What is being modeled?**
- A **multi-layer graph** $G = (V, E)$ where:
- **Nodes (V)** = people, households, devices, organizations, locations
- **Edges (E)** = relationships/interactions (social, financial, mobility, digital)





Layers of a real-world population graph

Data sources (at scale)
Telecom metadata (CDRs) Banking / payments
Public records (census, land, IDs), Mobility (GPS, transit cards)
Online platforms, IoT / sensors
Scale: **millions → billions of nodes, trillions of edges**

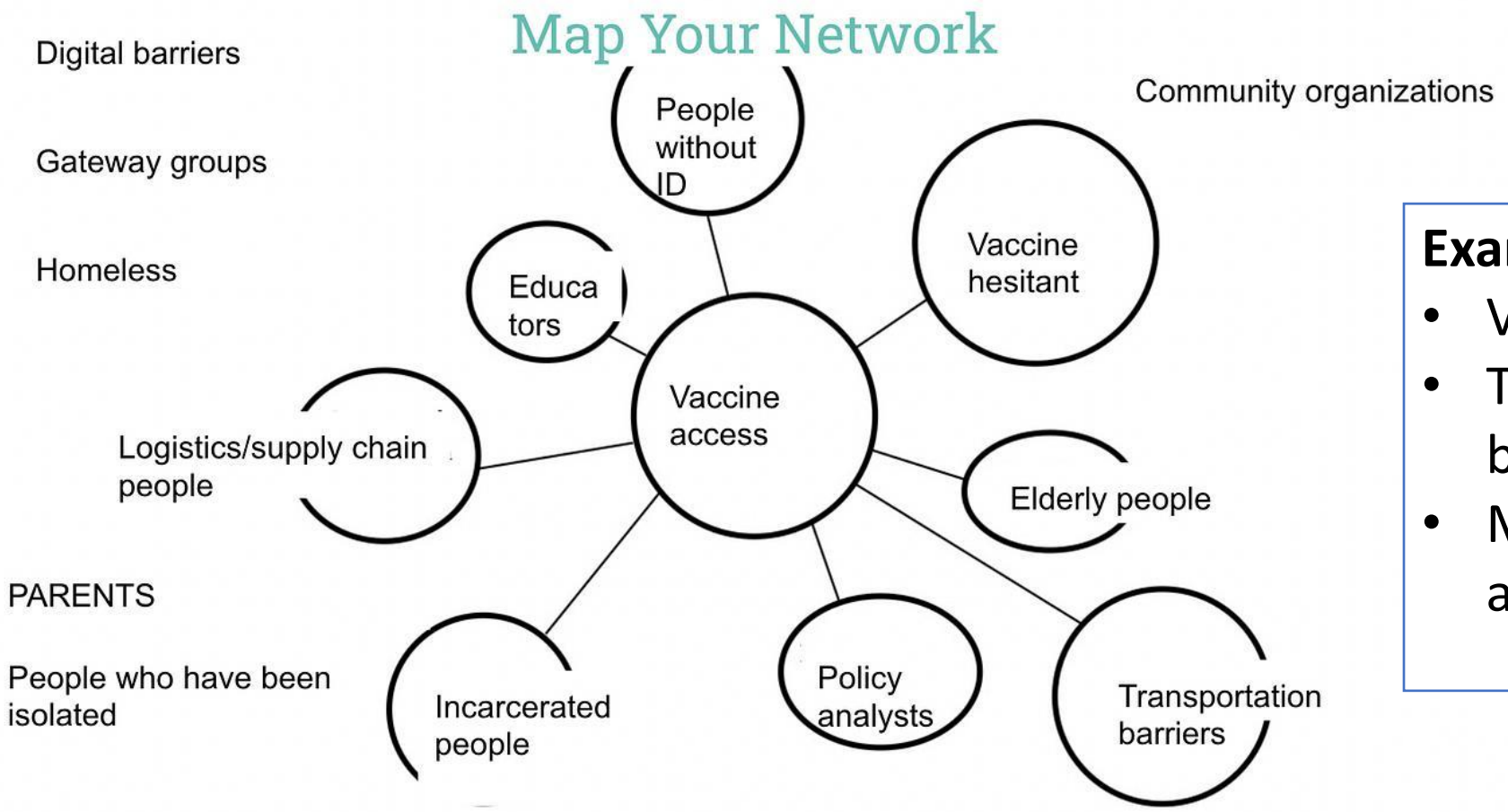
Layer	Nodes	Edges
Social	Individuals	friendships, calls, messages
Mobility	People, locations	travel paths, co-location
Financial	Accounts, merchants	transactions
Health	Patients, providers	visits, diagnoses
Digital	Devices, identities	logins, sessions

Core graph analytics

1. Centrality (influence detection) • Degree centrality • Betweenness centrality • Eigenvector/PageRank <u>Identifies</u> **key actors / hubs**	**2. Community detection** • Louvain / Leiden algorithms • Detect **clusters (communities)** <u>Example:</u> • Informal settlements • Online echo chambers • Fraud rings	**3. Link prediction** • Predict missing edges • Infer hidden relationships <u>Used in</u>: • Identity resolution (ties to FONDUE) • Fraud detection	**4. Edge weighting** • Incorporate **Deep Link Entropy** <u>Distinguish</u>: • Weak vs strong signals • Noise vs evidence

Network Interventions (the actionable layer)

- Once you understand the graph, you **change it** to achieve goals.



Examples

- Vaccinate "super-spreaders"
- Target influencers for behavior change
- Monitor high-risk financial actors

A. Edge-targeted interventions **Remove or weaken harmful edges** **Strengthen beneficial ones** **Examples** • **Limit fraudulent transaction channels** • **Introduce friction in risky interactions** • **Strengthen trusted identity links**	**C. Structural interventions** **Rewire network topology** **Examples** • **Add bridges between isolated communities** • **Break tightly-knit malicious clusters**
B. Structural interventions Rewire network topology **Examples** • Add bridges between isolated communities • Break tightly-knit malicious clusters	**D. Information interventions** Inject signals into network **Examples** • Public awareness campaigns • Misinformation correction via trusted nodes
E. Incentive-based interventions Change behavior via economic signals **Examples** Subsidies, Penalties, Dynamic pricing …	

Use cases

• **Urban Planning** • Optimize transport networks • Identify underserved communities • Improve service delivery	**Public Health (Epidemiology)** • Map contact networks • Identify super-spreaders • Target vaccination/quarantine Used in modeling outbreaks like **COVID-19**	
Financial Fraud & AML • Detect transaction clusters • Identify mule networks • Break illicit flows	**Digital Governance & Identity** • Build population-scale identity graphs • Deduplicate citizens (FONDUE-like systems) • Improve welfare targeting	**Information Security** • Map attacker infrastructure • Disrupt botnets and phishing networks

Summary

- Population-scale network mapping converts societies into **computable graphs**, and network interventions turn those insights into **targeted, high-leverage actions**.

Information Diffusion via Explosion / Shock-Wave Theory

- **Conceptual mapping**
- Treat a piece of information (post, rumor, campaign) as an **energy release** at time $t_0$that propagates through a network like a **blast wave**.
- **Source node(s)** → detonation point
- **Edges** → transmission medium
- **Exposure/reshare** → wavefront propagation
- **Engagement intensity** → pressure/energy

NGO Interest Groups on Facebook

- **What is an “interest group”?**
- A subgraph of users/pages uniting around:
- Causes (health, environment, rights)
- Campaigns (fundraisers, petitions)
- Shared identity or locality

Interventions for NGO campaigns

- **A. Maximize reach (benign campaigns)**
- Seed **top-k hubs + bridge nodes**
- Time posts to minimize λ(event alignment)
- Reinforce **low-entropy edges** (trusted endorsers)

- **B. Contain misinformation**
- Target **bridge nodes** (cut cross-community propagation)
- Inject corrective info at **high-centrality nodes**
- Increase friction on high-risk edges (reduce w_{ji})

Risks & biases

- **Echo chambers** → overestimation of reach
- **Algorithmic amplification** → non-stationary w_{ji}
- **Data gaps** (private groups, encrypted sharing)
- **Adversarial behavior** (coordinated inauthentic activity)

Summary

- **<u>Synthesis</u>**
- Shock-wave theory provides a **physics-inspired prior**: early energy, radial decay, thresholds.
- Network models provide **granularity**: topology, edge weights, community structure.
- For NGO activity on Facebook, effectiveness hinges on **hubs + bridges + timing**, with **edge quality (entropy)** governing credibility and cross-community spread.